Generative AI and Student Work Authenticity: Examining the Tension Between Strategic Reasoning and Academic Integrity in Higher Education Business Education

Authors:

Dr. Aida Mehrad

Ayan Chakraborty

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Abstract

Generative artificial intelligence (GenAI) has unsettled conventional assumptions about what makes assessed student work authentic. This paper examines that problem in higher education, business education and management programs where assignments are expected to reveal not only subject knowledge expertise but also contextual reasoning, theoretical application and strategic judgement. It develops a theory-adapting conceptual framework supported by structured evidence synthesis and secondary analysis of published aggregate data. The framework links the mode of GenAI engagement to cognitive engagement, strategic synthesis and assessor authenticity confidence, while treating authenticity-warrant strength as an evidential condition and institutional guidance as a contextual influence on disclosure. The synthesis suggests that critical and strategically augmentative AI use is more consistent with visible reasoning than substitutive use. However, the available studies do not test the full pathway in a common participant-level dataset; the proposed relationships therefore remain theory-derived propositions rather than confirmed causal effects. This paper argues that authenticity should be judged through converging evidence of intellectual ownership—including process traceability, reasoning evidence, disclosure consistency and corroborating performance—rather than through AI-detection scores alone. This approach reframes authenticity as a justified assessment judgement and offers practical direction for assessment design, assessor literacy and institutional policy.

Keywords: generative artificial intelligence; student work authenticity; cognitive engagement; strategic synthesis; academic integrity; assessment validity; higher education

1. Introduction

1.1 Generative AI and the student-authenticity problem

The rapid spread of generative artificial intelligence (GenAI) is reshaping established conceptions of authentic student work in higher education, particularly in business and management education, where assessment is increasingly expected to demonstrate knowledge application, critical synthesis, contextual reasoning, and strategic judgment rather than the simple reproduction of information. Generative AI tools, including Claude, Open AI (Chat GPT), Gemini, and other large language models (LLMs) can generate, restructure, paraphrase, and refine written material rapidly, thereby altering the relationship between students, knowledge production, and assessment. Empirical evidence indicates that GenAI has already become embedded in student assessment practices. For example, research involving 337 Australian university students found that more than one-third had used chatbots to support assessment-related activities, while some students did not necessarily regard such use as constituting an academic-integrity violation, highlighting the need for clearer institutional guidance and reconsideration of conventional assessment practices (Xia et al., 2024). The central educational issue is therefore not simply whether students use AI, but how, why, and to what extent they incorporate AI into the production of assessed work. Within business and management education, GenAI may legitimately support brainstorming, language refinement, information exploration, and the generation of alternative perspectives, provided students retain responsibility for evaluating theoretical relevance, contextual appropriateness, and strategic implications. However, the same technology may also facilitate the outsourcing of substantial elements of analysis and judgment, potentially producing fluent and sophisticated prose without equivalent evidence that the intended learning outcomes have been achieved. Research focusing specifically on business education similarly identifies potential benefits of

GenAI, including knowledge acquisition, co-ideation, personalized learning, and cognitive augmentation, while also highlighting concerns about cognitive dependency, reliability, assessment integrity, and the preservation of human and strategic capabilities (Cambridge University Press, 2024). GenAI should therefore not be conceptualized as simply an educational threat or an educational solution; rather, its educational consequences depend substantially on the nature of student–AI interaction and the assessment environment in which that interaction occurs.

This conceptualization is particularly relevant to HR, Business and Strategy assignments, where the intellectual value of assessment lies not primarily in producing grammatically sophisticated prose but in students' ability to select relevant evidence, interpret organizational circumstances, apply theoretical concepts, and establish defensible relationships between organizational diagnosis and strategic recommendations. Strategic synthesis therefore represents a distinct form of performance from generic AI-generated prose. It requires students to integrate multiple sources of information into a coherent explanation of an organizational problem and to determine which theoretical and empirical insights are relevant to the specific context. For example, an HR or business strategy assignment addressing employee retention in a small or medium-sized enterprise (SME) may require students to integrate labor-market conditions, organizational capabilities, employee motivation, leadership practices, and resource constraints before developing an appropriate intervention. Although GenAI can contribute to each stage of this process, its contribution does not in itself demonstrate learning. Research involving 118 postgraduate management students found that GenAI was used predominantly for writing, paraphrasing and rephrasing, whereas only a relatively small proportion of students demonstrated more advanced forms of AI-supported sensemaking examined in the study (Fischer et al., 2024). This finding suggests that the principal educational concern may extend beyond producing AI-generated text to the potential displacement of

higher-order cognitive activity by relatively low-level textual assistance. Generic AI-generated prose may appear academically competent while remaining limited in contextual specificity; for example, it may provide a theoretically accurate explanation of AMO theory, employee engagement, or strategic HRM without showing why a particular theoretical perspective fits a particular organizational problem. Strategic synthesis, by contrast, is demonstrated through contextualized reasoning in which students evaluate AI-generated possibilities, identify inappropriate or unsupported assumptions, and adapt relevant insights to the organization's size, financial resources, workforce characteristics and structural conditions. The intellectual contribution therefore lies not necessarily in producing the initial text but in the student's capacity to evaluate, transform, and strategically apply the information generated by AI.

1.2 From AI detection to authenticity assessment

The increasing use of GenAI also extends the traditional debate around plagiarism and academic integrity to broader questions of authorship, learning, and authenticity. Conventional similarity-detection systems primarily identify textual overlap with existing sources, whereas AI-writing detection systems attempt to estimate whether linguistic patterns match machine-generated text. Importantly, AI detection is probabilistic and should not be treated as proof of authorship or academic misconduct. Turnitin's guidance acknowledges the possibility of false positives and states that AI-writing detection results should not be the sole basis for adverse action against a student (Turnitin, 2026). Consequently, an AI-detection indicator, such as 26%, should not be interpreted as a universal threshold separating authentic from inauthentic work. Instead, such indicators may provide a starting point for further contextual investigation. An overreliance on detection technologies risks shifting assessment away from evaluating what students understand and can do toward assessing whether their linguistic output resembles a particular conception of human-authored writing. This may conflate three analytically distinct constructs: AI usage, academic misconduct, and learning. A student may use AI transparently

while demonstrating substantial independent learning, whereas a submission that does not trigger an AI detector may nevertheless provide limited evidence of meaningful disciplinary understanding. Consequently, recent scholarship advocates moving beyond a binary distinction between human-written and AI-generated work towards assessment approaches that emphasize self-regulated learning, responsible AI use, academic integrity, and the visibility of students' reasoning processes (Lodge et al., 2024). Related research on assessment and learning outcomes similarly identifies a movement from conventional assessment practices toward more innovative and GenAI-informed approaches (Australian Journal of Educational Technology [AJET], 2024). Within this emerging perspective, authenticity can be understood less as the complete absence of technological assistance and more as the demonstrable ownership of reasoning, judgment and learning.

1.3 Research gap

Despite the rapid expansion of research on GenAI in higher education, several important gaps remain. First, much of the existing literature focuses on AI through broad concerns about academic integrity, assessment, and technology adoption rather than examining how AI affects the quality and nature of strategic reasoning in business management, strategy, and Human Resources assignments. Recent scholarship on GenAI in business education similarly indicates that systematic understanding of its academic and organizational implications remains underdeveloped (Cambridge University Press, 2024). Second, existing research often conceptualizes AI use as a relatively binary phenomenon, concealing important distinctions between brainstorming, information seeking, critique, translation, editing, paraphrasing, and wholesale content generation. Research involving management students illustrates this problem: reported AI use may concentrate on lower-level writing functions rather than advanced forms of sensemaking (Fischer et al., 2024). Third, AI-detection percentages may become problematic proxies for authenticity when interpreted without adequate consideration

of the underlying learning process. Turnitin's acknowledgment of potential false positives reinforces the importance of contextual evaluation rather than treating an AI-detection score as conclusive evidence of authorship or misconduct (Turnitin, 2026). Accordingly, this study conceptualizes student work authenticity as a multidimensional construct encompassing authorship, disciplinary understanding, contextual specificity, strategic reasoning, and transparent AI use. The central analytical distinction is between AI functioning as a cognitive scaffold and AI functioning as a substitute for student reasoning. The proposed conceptual relationship can therefore be represented as: AI access → mode of AI use → student cognitive engagement → strategic synthesis → observable assessment performance, with ability, motivation, and opportunity shaping the relationship between AI access and cognitive engagement, while assessment design and institutional policy influence the conditions within which students make decisions about AI use. This framework shifts the research focus from the relatively narrow question of whether a student used AI toward the more educationally significant question of whether AI-assisted work demonstrates the student's capacity to evaluate evidence, apply Management or HR theory, interpret organizational context, and construct defensible strategic judgments. Such an approach provides a stronger foundation for qualitative investigation and comparative analysis of student work while connecting the study of AI adoption in education with the broader organizational literature on technology-enabled adaptation: the critical issue is not simply the presence of technology, but how human capabilities, motivations, opportunities, and institutional practices evolve in response to it.

1.4 Research aim and objectives

This paper aims to develop and evaluate an evidence-based framework for judging the authenticity of student work in GenAI-enabled business education. It addresses the following objectives:

- Distinguish critical and strategically augmentative GenAI engagement from forms of use that substitute for student reasoning.
- Explain how cognitive engagement and strategic synthesis may connect GenAI engagement to assessor confidence in the authenticity of student work.
- Develop the Authenticity Warrant as a convergent evidential basis for attributing demonstrated capability to the student.
- Assess the extent to which published empirical and theoretical evidence supports the proposed relationships.
- Identify implications for assessment design, assessor literacy and institutional AI guidance.

1.5 Research questions and propositions

The research paper is guided by three fundamental questions:

- Q1: How does the mode of GenAI engagement shape students' cognitive engagement and strategic synthesis in assessed business education work?
- Q2: What evidence enables assessors to attribute the reasoning and capability demonstrated in GenAI-assisted work to the student?
- Q3: How do assessment conditions and institutional AI guidance influence authenticity judgements and disclosure consistency?

Seven theory-derived propositions are developed to address these questions. They are labelled H1–H7 for continuity with the conceptual model, although the present evidence-synthesis design does not test them as statistical hypotheses in a common individual-level dataset.

- **H1:** More critical and strategically augmentative modes of GenAI engagement are positively associated with cognitive engagement.
- **H2:** Cognitive engagement is positively associated with strategic synthesis.

- **H3:** Strategic synthesis is positively associated with authenticity confidence.
- **H4:** More critical and strategically augmentative modes of GenAI engagement are positively associated with authenticity confidence.
- **H5:** Cognitive engagement and strategic synthesis serially mediate the relationship between the mode of GenAI engagement and authenticity confidence.
- **H6:** Authenticity Warrant strength positively moderates the relationship between strategic synthesis and authenticity confidence, such that the relationship is stronger when authenticity evidence converges.
- **H7:** Greater institutional AI-guidance clarity is positively associated with disclosure consistency.

where X is the mode of GenAI engagement, M1 is cognitive engagement, M2 is strategic synthesis, and Y is authenticity confidence. Authenticity Warrant strength W moderates the relationship between M2 and Y.

Because the present paper does not test these relationships using a common individual-level dataset, H1–H7 should be described as **theory-derived propositions** rather than proven hypotheses. Existing research can motivate and support the proposed relationships, while empirical validation would require matched measurements of all constructs from the same participants.

1.6 Contribution of the study

The paper makes three contributions. Theoretically, it separates the quality of a submitted product from confidence that the capability represented by that product belongs to the student. Methodologically, it introduces the Authenticity Warrant as a way of combining process traceability, reasoning evidence, disclosure consistency and corroborating performance without treating any single indicator as conclusive. Practically, it redirects attention from

attempts to detect AI-generated language towards assessment arrangements that make intellectual ownership visible and open to fair evaluation.

2. Literature Review

2.1 Generative AI in higher education

The rapid integration of generative artificial intelligence (GenAI) into higher education is transforming how students access information, develop arguments, and produce assessed work. This transformation is particularly significant within business education, where assessments in human resource management (HRM) and strategic management are designed not merely to evaluate the reproduction of disciplinary knowledge but also students' ability to diagnose organizational problems, integrate theoretical perspectives, evaluate evidence, and formulate contextually appropriate strategic recommendations. The increasing availability of GenAI models therefore presents both opportunities to enhance learning and significant challenges for academic integrity, assessment validity, and the extent to which conventional assessment practices can continue to provide reliable evidence of student learning (Xia et al., 2024). Rather than a straightforward technological disruption, GenAI raises a more complex question about the relationship between technological assistance and academic authenticity. The critical issue is not simply whether students use AI, but whether such use supports the development of students' analytical and strategic capabilities or substitutes for the cognitive processes that assessment is intended to evaluate. This distinction matters in business and HRM education, where the educational value of an HR strategy assignment lies in the student's ability to interpret an organizational problem, apply relevant theoretical concepts, integrate contextual evidence, and develop a defensible strategic response. Evidence from Fischer et al. (2024), based on 118 postgraduate management students, indicates that GenAI was used predominantly for writing, paraphrasing and rephrasing, while only a relatively small proportion of students

demonstrated more advanced forms of AI-supported sensemaking. This finding raises concerns that GenAI's accessibility may encourage the displacement of higher-order cognitive engagement by low-level textual assistance, potentially producing academically fluent outputs without equivalent evidence of independent strategic reasoning. The challenge is further complicated by the emergence of AI-writing detection systems, which attempt to identify linguistic patterns associated with AI-generated or AI-assisted text. Although such technologies may provide useful indicators for further academic inquiry, an AI-detection percentage should not be interpreted as a definitive measure of either authorship or academic misconduct. Turnitin (2026), for example, acknowledges that its AI-writing detection system may produce false positives and advises that AI-detection results should not be used as the sole basis for adverse action against a student. Consequently, a numerical indicator such as 26% should be understood as a signal requiring contextual consideration rather than as an established threshold separating authentic from inauthentic student work. This distinction matters because AI use, academic misconduct, and learning are conceptually separate constructs: a student may use AI transparently while demonstrating substantial independent learning, whereas a submission that does not trigger an AI detector may still provide limited evidence of meaningful strategic understanding. The emerging literature on GenAI and assessment consequently advocates shifting away from assessment models that focus predominantly on the final written product and toward approaches that make students' reasoning, reflection, and learning processes more visible. Xia et al. (2024), for example, argue for assessment practices that promote self-regulated learning, responsible GenAI use, and academic integrity in response to the changing educational environment. Nevertheless, a significant research gap remains concerning the specific relationship between AI use and strategic synthesis within Business, Management and Human Resource (HR) education. Existing research has increasingly examined patterns of AI adoption, academic integrity, and

assessment redesign, but it has paid comparatively less attention to distinguishing between legitimate AI-assisted synthesis and generic AI-generated business prose. This distinction matters because strategic synthesis requires students to transform information into contextually meaningful judgment rather than merely reproduce coherent language. In an HR strategy assignment, for instance, authentic performance may be demonstrated when a student evaluates AI-generated suggestions against the characteristics of a particular organization, identifies limitations in generic recommendations, applies relevant HR theory and develops a strategic response that reflects the organization's resources, workforce and operating environment. The intellectual contribution therefore resides not necessarily in producing every sentence independently, but in demonstrating ownership of the reasoning through which evidence, theory and contextual information are evaluated and integrated. This perspective is also particularly relevant to small and medium-sized enterprises (SMEs), where constrained financial, technological and human resources can make contextualized strategic judgment especially important. Generic AI-generated recommendations may fail to account for these constraints, whereas authentic strategic synthesis requires students to determine which recommendations are feasible and theoretically defensible within the specific organizational context. Furthermore, authenticity extends beyond the submitted text to the psychological and institutional conditions surrounding AI use. Where students perceive that any use of GenAI may automatically be interpreted as misconduct, they may be reluctant to disclose their use of AI, thereby creating a tension between institutional expectations for transparency and students perceptions of academic risk. A more developmental approach would therefore distinguish between AI as a cognitive scaffold and AI as a substitute for student reasoning, while encouraging students to demonstrate how they have evaluated, modified, accepted or rejected AI-generated material. Accordingly, this study investigates how students use GenAI when completing HR strategy assignments and how educators can evaluate the authenticity of student

work without reducing assessment to AI-detection scores. Authenticity is conceptualized not as the complete absence of technological assistance, but as evidence of student ownership of reasoning, disciplinary and contextual understanding, theoretical application, critical evaluation and strategic judgment. This conceptualization provides a basis for examining whether GenAI functions primarily as a tool that enhances students' cognitive engagement or as a mechanism that displaces key elements of strategic reasoning, thereby providing a more nuanced foundation for evaluating authentic learning in AI-enabled business education.

This distinction is closely connected to emerging conceptualizations of AI literacy. Ng et al. (2024) identify six dimensions of AI literacy: recognizing AI, understanding AI, using and applying AI, evaluating AI, creating with AI, and ethically navigating AI. From an assessment perspective, the capacity to evaluate and ethically navigate AI-generated information may therefore be as important as the ability to formulate effective prompts. Authentic performance in an AI-enabled educational environment may increasingly require students to interrogate AI outputs, identify inaccuracies and limitations, compare alternative explanations, and transform generic information into contextually appropriate knowledge. The Ability–Motivation–Opportunity (AMO) framework (Vroom, 1964; Blumberg & Pringle, 1982) provides a useful theoretical lens for explaining why students may engage with GenAI in different ways. AMO theory proposes that performance is shaped by individuals' ability, motivation, and opportunity to perform, and a systematic review of quantitative HRM research shows the framework's continued relevance while also highlighting variation in how researchers conceptualize and measure its components (Marin-Garcia et al., 2023). Applied to AI-supported assessment, ability may encompass disciplinary knowledge, analytical capability and AI literacy, because students require sufficient subject knowledge to determine whether AI-generated explanations are theoretically sound and contextually appropriate. Motivation concerns the reasons students engage with the assessment task: students motivated primarily by efficiency may use AI to

minimize cognitive effort, whereas those motivated by developing strategic competence may use AI to challenge assumptions, explore alternatives, and obtain feedback. Opportunity concerns the conditions within which students can exercise their ability and motivation, including assessment requirements, institutional AI policies, access to AI technologies, lecturer expectations and opportunities for formative feedback. This framework therefore suggests that problematic AI use should not be understood exclusively as an individual behavioral issue, because assessment systems and institutional practices can create incentives for particular patterns of AI use. Where assessment places disproportionate emphasis on polished written output, students may have stronger incentives to outsource textual production; conversely, assessments requiring process evidence, contextual justification, reflective explanation and oral defense may create stronger opportunities for students to retain ownership of strategic reasoning.

2.2 Student work authenticity

The earlier framework concluded with observable assessment performance. The revised framework goes further by distinguishing the quality of the assessed product from confidence in its ownership. A strategically coherent submission may reflect substantial student reasoning, substantial AI substitution, or a combination of the two. The final product alone does not uniquely identify which process occurred.

The dependent variable is therefore **student-work authenticity confidence**, defined as the assessor's degree of justified confidence that the reasoning, decisions, and capabilities demonstrated in a submission can credibly be attributed to the student.

This definition treats authenticity as a judgement supported by evidence rather than an intrinsic property of a document. A student may authentically produce weak work, while an excellent submission may offer insufficient evidence of student ownership. Similarly, compliance and authenticity should remain distinct. A student may follow disclosure rules but rely heavily on

AI reasoning, or breach a disclosure requirement while independently undertaking much of the intellectual work. Both issues matter, but they require different judgements.

2.3 Academic integrity and AI disclosure

Social Exchange Theory (SET) originated with Homans' (1958) conceptualization of social behaviour as exchange and was subsequently extended by Thibaut and Kelley (1959), Blau (1964), and Emerson (1976). Within organizational research, SET explains how favourable treatment, support, and valued resources can create reciprocal obligations that encourage individuals to respond with trust, commitment, engagement, and constructive behaviour (Cropanzano & Mitchell, 2005). contributes a relational and institutional perspective. SET proposes that social relationships develop through exchanges involving benefits, obligations, reciprocity, fairness, and trust. In higher education, students are expected to participate honestly in learning and assessment, while educators and institutions are expected to provide fair judgement, transparent requirements, and appropriate support.

GenAI complicates this exchange by introducing uncertainty about what constitutes legitimate assistance. If institutional expectations are unclear or inconsistently applied, students may be unsure whether particular uses should be disclosed. If disclosure is assumed to signal misconduct, even students who use AI permissibly may be reluctant to provide a complete account. Conversely, clear boundaries and proportionate responses may strengthen trust and make accurate disclosure more likely.

SET therefore supports the proposed contextual relationship between **institutional AI-guidance clarity** and **disclosure consistency**. Guidance clarity concerns whether permitted uses, prohibited uses, disclosure requirements, and consequences are communicated in understandable and consistent terms. Disclosure consistency concerns whether a student's account of AI use aligns with the process and product evidence available.

Disclosure should not, however, be equated with authenticity. A student may accurately disclose extensive dependence on GenAI, which would demonstrate transparency but not necessarily cognitive ownership. Another student may provide an incomplete disclosure because of ambiguity or misunderstanding while still having undertaken much of the reasoning independently. SET helps explain the relational conditions under which disclosure occurs; it does not independently establish whether the reasoning represented in the submission belongs to the student.

Trust in educators must also be distinguished from trust in automation. The former concerns expectations of fair and proportionate institutional treatment. The latter concerns reliance on the quality of AI-generated information. Confusing these two forms of trust would obscure the different mechanisms involved.

2.4 Cognitive engagement

Cognitive offloading refers to the use of external actions or resources to change the information-processing demands of a task (Risko & Gilbert, 2016). Offloading is not inherently harmful. Students regularly use calculators, notes, databases, and other resources to reduce routine demands. The educational issue is whether offloading frees capacity for higher-order reasoning or displaces the reasoning the assessment is intended to measure.

This distinction gives greater precision to the concepts of cognitive scaffolding and substitution. Scaffolding occurs when assistance supports reasoning that the student continues to undertake and evaluate. Substitution occurs when the student delegates assessed reasoning and accepts the result with insufficient scrutiny.

For example, using GenAI to organise research notes may allow a student to focus on comparing explanations for employee turnover. In contrast, accepting an AI-generated diagnosis without examining its assumptions may delegate the central analytical task. A student might also begin with a generated draft but substantially reconstruct it after verifying evidence

and reconsidering the argument. The relevant question is therefore not how much language originated from AI, but which cognitive functions were delegated, retained, or recovered through revision.

Self-regulated learning explains how students plan, monitor, and evaluate their activities. Zimmerman (2002) described self-regulation as a cyclical process involving forethought, performance, and self-reflection. Applied to GenAI-assisted assessment, students may define the problem and establish goals before using AI, monitor the credibility of generated material during the task, and evaluate how their position changed afterwards.

These activities provide theoretically relevant evidence of cognitive engagement. Indicators may include checking sources, questioning assumptions, comparing alternatives, correcting inaccuracies, documenting consequential revisions, and explaining why particular recommendations were accepted or rejected. Receiving a generated critique does not itself demonstrate self-regulation. The student must decide whether the critique is valid and how it should affect the developing argument.

Lee and See's (2004) account of trust in automation provides a related explanation. Appropriate reliance requires users to calibrate their trust according to the system's capabilities and limitations. In GenAI-assisted work, confidence in a fluent response may encourage acceptance, whereas calibrated reliance requires verification and awareness of uncertainty. Appropriate reliance is therefore relevant to cognitive engagement, although it is distinct from the institutional trust addressed through SET.

The ICAP framework differentiates passive, active, constructive, and interactive engagement (Chi & Wylie, 2014). Passive engagement involves receiving information, active engagement involves manipulating it, constructive engagement involves generating ideas or inferences beyond the presented material, and interactive engagement involves mutually generative dialogue.

Applying ICAP to GenAI requires care. A long conversation with an AI system is not necessarily evidence of interactive learning, and extensive editing does not necessarily demonstrate constructive engagement. More convincing evidence may include identifying contradictions, producing an explanation not supplied by the system, reconciling theoretical perspectives, evaluating evidence, or revising a position in response to a substantive challenge.

In the proposed framework, **cognitive engagement is the first serial mediator**. It refers to the degree to which students actively interpret, verify, evaluate, monitor, revise, or reject AI-generated material. It explains how the mode of GenAI engagement may influence the reasoning activities retained by the student.

The construct should remain distinct from engagement frequency. A student who submits many prompts may nevertheless engage superficially, while another student may use AI briefly but evaluate the response carefully. Cognitive engagement concerns the quality of reasoning, not simply the quantity of interaction.

2.5 Strategic synthesis

Cognitive engagement is necessary but may not be sufficient for strategic synthesis. A student may identify flaws in an AI-generated answer without integrating theory, evidence, contextual information, and competing priorities into a coherent recommendation. Strategic synthesis is therefore treated as the second serial mediator.

Strategic synthesis refers to the integration of disciplinary theory, empirical or organisational evidence, contextual constraints, stakeholder interests, and alternative courses of action into a defensible judgement. In HRM education, it may involve reconciling organisational performance with employee wellbeing, legal obligations, ethical considerations, implementation constraints, and longer-term consequences.

This construct differs from cognitive engagement in its focus on integration and decision-making. Cognitive engagement concerns the activities through which information is interpreted

and evaluated. Strategic synthesis concerns whether those activities produce a coherent and contextually justified position. The distinction supports a serial process:

Mode of GenAI engagement → Cognitive engagement → Strategic synthesis

Fischer et al. (2024) provide an empirical basis for distinguishing levels of sensemaking, but their aggregate findings should not be presented as evidence that the complete serial process has been validated. The study motivates the pathway; it does not estimate its mediation effects.

2.6 Authenticity evidence and warrants

The proposed **Authenticity Warrant** is the principal theoretical extension introduced by this study. It represents the strength and convergence of the evidence supporting the attribution of demonstrated capability to the student.

The warrant comprises four dimensions:

Process traceability: Evidence showing how the work developed, including selected prompts, drafts, revisions, source checks, and consequential decisions.

Reasoning evidence: The student's ability to explain assumptions, evaluate alternatives, justify revisions, and defend conclusions.

Disclosure consistency: The alignment between the student's account of GenAI use and other available process and product evidence.

Corroborating performance: Consistency between the submitted work and related demonstrations, such as an oral defence, supervised activity, reflective explanation, or transfer task.

No individual source of evidence is treated as conclusive. Prompt records can document activity without demonstrating understanding. A polished reflection may itself have been generated. An oral defence may be influenced by language proficiency or anxiety. Draft histories may be incomplete. The warrant therefore depends on the convergence of different indicators rather than on any single form of verification.

The Authenticity Warrant is modelled as a moderator. It does not cause cognitive engagement or create strategic synthesis. Instead, it affects whether observed synthesis provides a credible basis for attribution. When warrant strength is high, evidence of strategic synthesis should translate more strongly into authenticity confidence. When warrant strength is low, assessors may recognise apparent synthesis while remaining uncertain about whose capability it represents.

2.7 Assessor authenticity confidence

Authenticity confidence may draw on the quality of the final submission, the student's explanation of the reasoning, consistency with prior or supervised performance, and evidence showing how the work developed. It should not be inferred from AI-detection results alone. Product-level indicators may prompt further review, but they cannot independently establish whether the demonstrated capability belongs to the student.

2.8 Institutional AI-guidance clarity

Assessment design therefore critically determines whether GenAI enhances or undermines authentic learning. Traditional essay-based assessments are particularly vulnerable because the final written product can be generated, extensively revised, or stylistically transformed without necessarily revealing the cognitive processes behind it. Recent literature therefore advocates assessment approaches that make reasoning, reflection, and learning processes more visible (Lodge et al., 2024). Such approaches may require students to document selected AI interactions, explain how they evaluated AI-generated responses, identify inaccuracies or weaknesses in AI outputs, and justify which suggestions they retained, modified, or rejected. This approach transforms AI use from a largely hidden variable into an observable component of the learning process and may also support greater transparency by reducing the incentive for students to conceal legitimate forms of AI assistance. Such assessment practices are

particularly compatible with business and strategic management education, where judgment under uncertainty is central to professional practice. Students should therefore be evaluated not simply on their ability to produce a plausible answer but also on their capacity to evaluate alternatives, recognize contextual limitations, justify theoretical choices and defend strategic decisions.

Opportunity concerns the environmental conditions that enable, constrain, or encourage particular behaviours. In GenAI-assisted assessment, these conditions include access to technology, task design, permitted-use rules, disclosure requirements, feedback arrangements, assessment criteria, and the extent to which students are required to explain their reasoning.

Opportunity is particularly important because assessment design influences both student behaviour and evidential visibility. A task that rewards a polished final answer without requiring contextual justification or explanation may provide greater scope for cognitive substitution. By contrast, an assessment that requires students to document consequential decisions, explain rejected alternatives, verify sources, or defend a recommendation may encourage more deliberate engagement while producing stronger evidence of student ownership.

Xia et al. (2024) and Weng et al. (2024) reviewed the changing relationship between GenAI and assessment in higher education. Their reviews support the need to reconsider assessment design, learning outcomes, self-regulation, and academic-integrity practices in response to GenAI. Because these publications are scoping reviews, they should be used to establish the broader assessment context and identify research gaps rather than presented as direct tests of the relationships proposed in this study.

Within the conceptual model, opportunity has two functions. First, it influences which modes of GenAI engagement are permitted and encouraged. Second, it influences whether the assessment generates sufficient evidence for an assessor to distinguish student-owned

reasoning from AI-supported production. Opportunity therefore shapes both the development and the visibility of authenticity.

SDT also informs the conditions surrounding disclosure, although it should not be used as the principal theory of institutional trust. Students who experience assessment rules as highly controlling or punitive may focus on avoiding sanctions rather than engaging openly with expectations. Clear and proportionate guidance may support more autonomous regulation by helping students understand not only what is permitted but why particular boundaries exist. This relationship is theoretically plausible, but psychological safety should not be claimed as an empirically measured construct unless the study includes an appropriate validated measure.

2.9 Theoretical foundations

2.9.1 Ability–Motivation–Opportunity framework

The educational consequences of generative artificial intelligence (GenAI) cannot be explained by access to technology alone. Students with access to the same platform may use it in fundamentally different ways. One student may accept a generated answer with little scrutiny, another may use the system to clarify unfamiliar material, and a third may test an independently developed argument against alternative interpretations. Although all three students have "used AI," the intellectual work retained by each student differs considerably. Theoretical attention must therefore move beyond the presence or frequency of GenAI use and examine the purposes for which the technology is used, the reasoning activities retained by the student, and the evidence available to establish ownership of the resulting work.

This study draws principally on the **Ability–Motivation–Opportunity** framework, Self-Determination Theory, and Social Exchange Theory. These perspectives explain different but complementary aspects of AI-assisted assessment. AMO provides the organising structure by explaining how students' capabilities, motivations, and opportunities shape their behaviour.

Self-Determination Theory clarifies why students may engage with GenAI as a learning resource or depend on it as a substitute for effort. Social Exchange Theory explains how institutional expectations, fairness, trust, and reciprocity influence disclosure and participation in verification practices.

These theories describe the conditions surrounding the central conceptual pathway rather than functioning as competing explanations. The core pathway proposes that the **mode of GenAI engagement** influences **student-work authenticity confidence** through the serial mechanisms of **cognitive engagement** and **strategic synthesis**. The proposed **Authenticity Warrant** moderates the final relationship by determining whether apparent strategic synthesis can credibly be attributed to the student. This distinction is important: the quality of a submission and confidence in its ownership are related, but they are not the same construct.

The AMO framework proposes that behaviour and performance depend on whether individuals possess the ability to act, are motivated to act, and have an opportunity to act. Within human resource management research, AMO has been used extensively to explain how organisational practices influence individual and organisational outcomes. Bos-Nehles et al. (2023), reviewing the conceptualisation and measurement of AMO in HRM research, emphasised that ability, motivation, and opportunity should not automatically be treated as interchangeable or additive. Their relationships may vary according to the context and outcome under investigation.

Applied to GenAI-assisted assessment, AMO helps explain why similar access conditions may lead to different forms of student engagement. Students differ in their disciplinary knowledge, analytical skills, AI literacy, motivation to learn, performance orientation, time constraints, and understanding of institutional rules. Assessment design also varies in the extent to which it permits GenAI use, requires disclosure, makes reasoning visible, and provides opportunities

for feedback. These conditions shape both how GenAI is used and whether students retain ownership of the reasoning represented in their work.

In the revised framework, AMO does not replace the focal independent variable. Instead, ability, motivation, and opportunity are treated as antecedent and contextual conditions that help explain variation in the **mode of GenAI engagement**. The mode of engagement remains the central explanatory variable because it captures what the student and the technology actually do within the assessment process.

Ability includes disciplinary knowledge, analytical capability, self-regulatory competence, and AI literacy. A student needs sufficient disciplinary understanding to evaluate whether an AI-generated claim is accurate, relevant, and appropriate to the problem being examined. Technical familiarity with an AI interface is insufficient if the student cannot identify a theoretically weak explanation, fabricated source, unjustified assumption, or contextually inappropriate recommendation.

This issue is particularly relevant in business management and HR education. A GenAI system may generate a plausible explanation for employee turnover or recommend an apparently coherent performance-management intervention. A student with stronger HRM knowledge may compare the response with relevant theory, identify omitted stakeholder interests, question causal assumptions, and assess whether the recommendation suits the organisational context. A student with weaker disciplinary knowledge may be more likely to accept the same response because it is fluent and professionally presented.

AI literacy provides another dimension of ability. Lintner's (2024) systematic review demonstrates that AI literacy has been conceptualised through multiple dimensions rather than as a single technical skill. In the context of this study, relevant capabilities include understanding what GenAI can and cannot do, recognising uncertainty and hallucination risks, verifying sources, and deciding when independent human judgement is required. AI literacy

consequently supports critical engagement, but it does not guarantee it. A highly capable student may still choose to delegate substantial reasoning to save time or improve a grade.

Ability is therefore expected to influence whether students can progress from simple generation or rephrasing to evaluative and strategically augmentative forms of engagement. It should not, however, be treated as direct proof of cognitive ownership. Capability describes what a student is able to do; ownership concerns what the student actually did in producing the assessed work.

Motivation concerns why students engage with an assessment and why they select particular forms of AI assistance. Students may use GenAI to develop their understanding, obtain feedback, compare alternatives, improve written communication, reduce uncertainty, save time, or maximise assessment performance. These motives are not mutually exclusive. A student can be committed to learning while also experiencing time pressure, and a performance-oriented student may still engage critically with AI output.

The distinction matters because the same technological function can support different goals. Asking GenAI to critique a proposed HR strategy may be part of a reflective learning process, or it may be used to produce evaluative language that the student does not independently understand. Similarly, requesting a summary may help a student enter an unfamiliar topic, but it may also replace engagement with the source material. Motivation therefore helps explain why technically similar AI activities can involve different levels of cognitive ownership.

Fischer et al. (2024) provide a useful empirical point of departure by distinguishing forms of GenAI-supported sensemaking and comparing students' claims with assessor evaluations. Their study directs attention away from undifferentiated measures of AI use and toward differences in how the technology contributes to assessment work. It also shows why students' descriptions of their engagement cannot automatically be treated as equivalent to the engagement visible to assessors. The study does not, however, determine which perspective is objectively correct or establish that a particular mode of use causes stronger authenticity.

2.9.2 Self-Determination Theory

Self-Determination Theory (SDT) provides a complementary perspective for understanding differences in students' motivation to use GenAI. SDT conceptualizes motivation in relation to psychological needs including autonomy, competence and relatedness, and recent research has applied SDT-informed models to students' continued use of ChatGPT in higher education (ScienceDirect, 2024). GenAI may support autonomy by enabling students to explore ideas independently and may support perceived competence through immediate feedback, explanation and scaffolding. However, excessive reliance on AI may also undermine competence if students increasingly perceive themselves as unable to formulate, evaluate or communicate ideas without technological assistance. AI use should therefore not automatically be interpreted as evidence of an intention to deceive. Students may engage with AI because of time constraints, uncertainty about academic writing, performance pressures, or difficulty translating conceptual knowledge into coherent written arguments. Evidence from Australian university students further indicates that psychosocial factors are associated with patterns of AI use, reinforcing the importance of examining the broader conditions surrounding student behavior rather than reducing AI engagement to a simple academic-integrity violation (Xia et al., 2024). From an SDT perspective, the critical issue is therefore whether GenAI supports or substitutes for the development of students' competence and autonomous judgment.

Self-Determination Theory provides a more detailed explanation of the motivational processes underlying GenAI engagement. SDT proposes that motivation is influenced by the satisfaction of three psychological needs: autonomy, competence, and relatedness. It also distinguishes more autonomous forms of motivation from behaviour driven primarily by external control, reward, or avoidance of punishment (Ryan & Deci, 2020).

GenAI may support student autonomy by enabling independent exploration and allowing students to seek assistance without waiting for formal instruction. It may support perceived

competence by offering explanations, examples, and immediate feedback. It may also affect relatedness if students use AI because conventional academic support feels inaccessible or if educators integrate AI into a supportive learning environment. Annamalai et al. (2025) applied a modified SDT model to students' motivation for continued ChatGPT use, indicating the relevance of motivational needs to sustained engagement with the technology.

These benefits should nevertheless be interpreted cautiously. Perceived competence and demonstrated competence are not identical. A student may feel more capable because GenAI produces an immediate, coherent response, although the student may not have developed the knowledge required to evaluate or reproduce the reasoning independently. Conversely, structured AI assistance may strengthen competence when students use it to test ideas, identify weaknesses, seek alternative perspectives, and revise their own analysis.

SDT therefore helps distinguish **AI-supported learning** from **AI-dependent performance**. In AI-supported learning, the student remains responsible for interpreting, evaluating, and applying the information. In AI-dependent performance, important cognitive activities are increasingly delegated to the system. The distinction cannot be inferred solely from the length of the prompt history or the proportion of generated language. It depends on the intellectual decisions the student retains and can explain.

2.9.3 Affordance actualisation and engagement modes

Affordance actualisation distinguishes what a technology makes possible from what users actually do with those possibilities. Strong et al. (2014) argued that technological outcomes depend on the relationship among the features of an artefact, the goals of users, and the context in which action occurs. This distinction is useful for understanding why GenAI access does not produce uniform educational outcomes.

A GenAI platform may offer capabilities for drafting, summarising, translating, generating alternatives, critiquing arguments, and simulating stakeholder perspectives. Yet availability

does not determine which of these possibilities a student will enact. One student may request a complete strategy and adopt it with minimal modification. Another may submit an independently developed proposal to the system for criticism. A third may use the tool only to clarify terminology.

In the revised framework, **AI access is treated as a precondition rather than the focal independent variable**. Access establishes that particular forms of assistance are technically available. The focal independent variable is the **mode of GenAI engagement**, which captures the intellectual purpose of the interaction and the degree of evaluative control retained by the student.

Four modes are distinguished:

- **Substitutive generation**, in which generated material is accepted with limited scrutiny.
- **Interpretive use**, in which GenAI explains, reframes, summarises, or compares information.
- **Evaluative use**, in which the student questions, verifies, critiques, revises, or rejects the output.
- **Strategic augmentation**, in which evaluated AI contributions are selectively integrated into an independently justified decision.

These categories should not be interpreted as a mechanically progressive hierarchy. A student may move between modes during the same assignment, and apparently evaluative prompts may conceal weak independent judgement. The classification concerns the intellectual function performed in a particular part of the task.

3. Conceptual Framework

3.1 Overview of the proposed framework

The proposed framework explains how the **mode of GenAI engagement** may influence confidence in the authenticity of student work. It moves beyond whether students use GenAI and instead examines how they use it, what reasoning they retain, and whether the demonstrated capability can credibly be attributed to them. The model specifies one independent variable, two serial mediators, one dependent variable, and one moderator.

The framework proposes that the mode of GenAI engagement, ranging from substitutive generation to strategic augmentation, influences student-work authenticity confidence. Cognitive engagement is the first mediator, capturing interpretation, verification, critical evaluation, and revision of AI-generated material. Strategic synthesis is the second mediator, reflecting integration of theory, evidence, context, and trade-offs into defensible decisions. Authenticity Warrant strength moderates the final relationship through process traceability, reasoning evidence, disclosure consistency, and corroborating performance. Accordingly, the model predicts positive direct, serially mediated, and moderated relationships, while institutional guidance is expected to strengthen disclosure consistency. These are theory-derived propositions requiring future empirical validation with data

3.2 Construct definitions

Construct	Definition and conceptual boundary	Indicative evidence
Mode of GenAI engagement	How students use GenAI, ranging from substitutive generation to strategic augmentation.	Acceptance, interpretation, verification, critique, revision, rejection and contextual integration of AI output.
Cognitive engagement	The degree to which students interpret, verify, evaluate, monitor and revise AI-supported material.	Source checks, assumption testing, comparisons, substantive revisions and explained rejection of output.
Strategic synthesis	Integration of theory, evidence, context, stakeholder interests and alternatives into a defensible judgement.	Contextualised recommendations, explicit trade-offs, theoretical justification and feasible implementation choices.

Construct	Definition and conceptual boundary	Indicative evidence
Authenticity confidence	The assessor's justified confidence that the reasoning and capability demonstrated can be attributed to the student.	Consistency between the product, process evidence, explanation and related performance.
Authenticity-warrant strength	The strength and convergence of evidence supporting attribution of demonstrated capability to the student.	Process traceability, reasoning evidence, disclosure consistency and corroborating performance.
Disclosure consistency	Alignment between the student's account of GenAI use and other available evidence.	Concordance among declarations, process traces, drafts and explanations.
Institutional AI-guidance clarity	The extent to which acceptable use, disclosure and evidential expectations are understandable and consistent.	Clear rules, stable terminology, aligned module guidance and proportionate procedures.

3.3 Proposition development

3.3.1 GenAI engagement and cognitive engagement (H1)

H1: More critical and strategically augmentative modes of GenAI engagement are positively associated with cognitive engagement.

H1 is assessed primarily through Fischer et al.'s (2024) classification of GenAI-supported sensemaking and through literature on cognitive offloading, self-regulated learning, appropriate reliance, and cognitive engagement. Evidence is considered consistent with H1 when evaluative or augmentative use involves interpretation, source verification, assumption testing, critical evaluation, substantive revision, or rejection of unsuitable AI output.

Fischer et al. provide empirically relevant classifications but do not directly estimate the association specified in H1. Accordingly, the study can provide **indirect empirical and theoretical support**, but not definitive statistical confirmation.

3.3.2 Cognitive engagement and strategic synthesis (H2)

H2: Cognitive engagement is positively associated with strategic synthesis.

H2 is evaluated through theoretical pattern matching. Evidence of cognitive engagement—including interpretation, monitoring, verification, evaluation, and revision—is compared with evidence of strategic synthesis, including theory–evidence

integration, contextual judgement, evaluation of trade-offs, and defensible recommendations.

Evidence supports H2 when the literature shows that active evaluation contributes to integrated and contextually justified reasoning. A sophisticated final product without evidence of the student's reasoning is treated as inconclusive because product quality alone does not establish cognitive engagement or ownership.

3.3.3 Strategic synthesis and authenticity confidence (H3)

H3: Strategic synthesis is positively associated with authenticity confidence.

H3 is assessed by examining whether integrated, contextualised, and defensible reasoning provides a credible basis for attributing demonstrated capability to the student. Relevant evidence includes the student's ability to explain theoretical choices, justify trade-offs, defend recommendations, and apply the reasoning to a related problem.

Authenticity confidence is not directly measured by Fischer et al. or Xia et al. H3 is therefore evaluated mainly through authentic-assessment literature and the formal attribution argument developed in this study. It should be classified as an **original theoretically supported hypothesis requiring future empirical testing**.

3.3.4 Engagement mode and authenticity confidence (H4)

H4: More critical and strategically augmentative modes of GenAI engagement are positively associated with authenticity confidence.

H4 examines the direct relationship between the independent and dependent variables. Fischer et al.'s comparison of student claims and assessor classifications is used to determine whether engagement mode provides relevant information about the

ownership of assessed reasoning. However, the reported differences also demonstrate that student descriptions cannot independently establish authenticity.

Consequently, Fischer et al.'s evidence can motivate H4, but it cannot estimate the proposed direct effect. H4 should be classified as **indirectly supported but not directly tested**.

3.3.5 Serial mediation through cognitive engagement and strategic synthesis (H5)

H5: Cognitive engagement and strategic synthesis serially mediate the relationship between the mode of GenAI engagement and authenticity confidence.

H5 is assessed by examining the evidence for each component of the proposed sequence:

X→M1→M2→Y

where X represents GenAI engagement mode, M1 cognitive engagement, M2 strategic synthesis, and Y authenticity confidence.

The methodology first evaluates H1, H2, and H3 separately and then examines whether their combined evidence forms a coherent explanatory sequence. This process can establish theoretical plausibility but cannot calculate a serial indirect effect because the available studies do not provide matched individual-level measures or compatible covariance data. H5 is therefore assessed as an **integrated theoretical mechanism rather than a statistically tested mediation hypothesis**.

3.3.6 Moderating role of authenticity-evidence convergence (H6)

H6: Authenticity Warrant strength positively moderates the relationship between strategic synthesis and authenticity confidence, such that the relationship is stronger when authenticity evidence converges.

Turnitin's large-scale reporting is used to demonstrate the practical prevalence of AI-indicated writing, but detector classifications are not treated as verified measures of misconduct, engagement mode, or student ownership. Evidence that detection is less reliable for hybrid human–AI texts further support the argument that product-level indicators cannot independently establish authenticity.

The South African study combining Turnitin results with student reflection provides indirect evidence relevant to H6. Reported difficulty among some highly flagged students in credibly explaining their AI use illustrates the potential value of combining automated indicators with reasoning and disclosure evidence. Nevertheless, the study does not measure strategic synthesis, total Authenticity Warrant strength, or assessor authenticity confidence. It therefore provides **indirect support for evidence convergence**, not a statistical test of the moderation effect.

3.3.7 Institutional guidance and disclosure consistency (H7)

H7: Greater institutional AI-guidance clarity is positively associated with disclosure consistency.

H7 is assessed through Xia et al.'s (2024) review of GenAI and assessment, together with literature addressing assessment design, policy clarity, permitted-use rules, fairness, and disclosure expectations. Evidence is considered consistent with H7 when clearer institutional expectations improve students' understanding of acceptable AI use or encourage more transparent reporting.

Because Xia et al. provide review-level evidence rather than matched individual-level measurements of guidance clarity and disclosure consistency, H7 can receive **contextual and theoretical support**, but not causal confirmation.

3.4 Conceptual model

The revised framework proposes the following core pathway:

Mode of GenAI engagement → Cognitive engagement → Strategic synthesis → Authenticity confidence

Authenticity Warrant strength moderates the final path:

Strategic synthesis X Authenticity Warrant strength → Authenticity confidence

AMO, SDT, SET, and affordance actualisation explain the antecedent and contextual conditions surrounding the pathway. Ability influences whether students can evaluate generated information. Motivation influences why they choose particular forms of assistance. Opportunity influences which behaviours are encouraged and what evidence becomes visible. SDT explains the motivational quality of engagement, while SET explains the role of trust, fairness, and reciprocity in disclosure and participation. Affordance actualisation explains why access must be converted into situated use.

Figure 1 below explains the Integrated Conceptual Model that explains the mode of Gen AI engagement amongst students framework should not be interpreted as a validated causal model. Students may move repeatedly between generation, interpretation, evaluation, and synthesis. Feedback can return students to earlier stages, and assessment conditions may influence several relationships simultaneously. The arrows represent theoretically proposed relationships selected for analysis rather than a complete representation of every possible interaction.

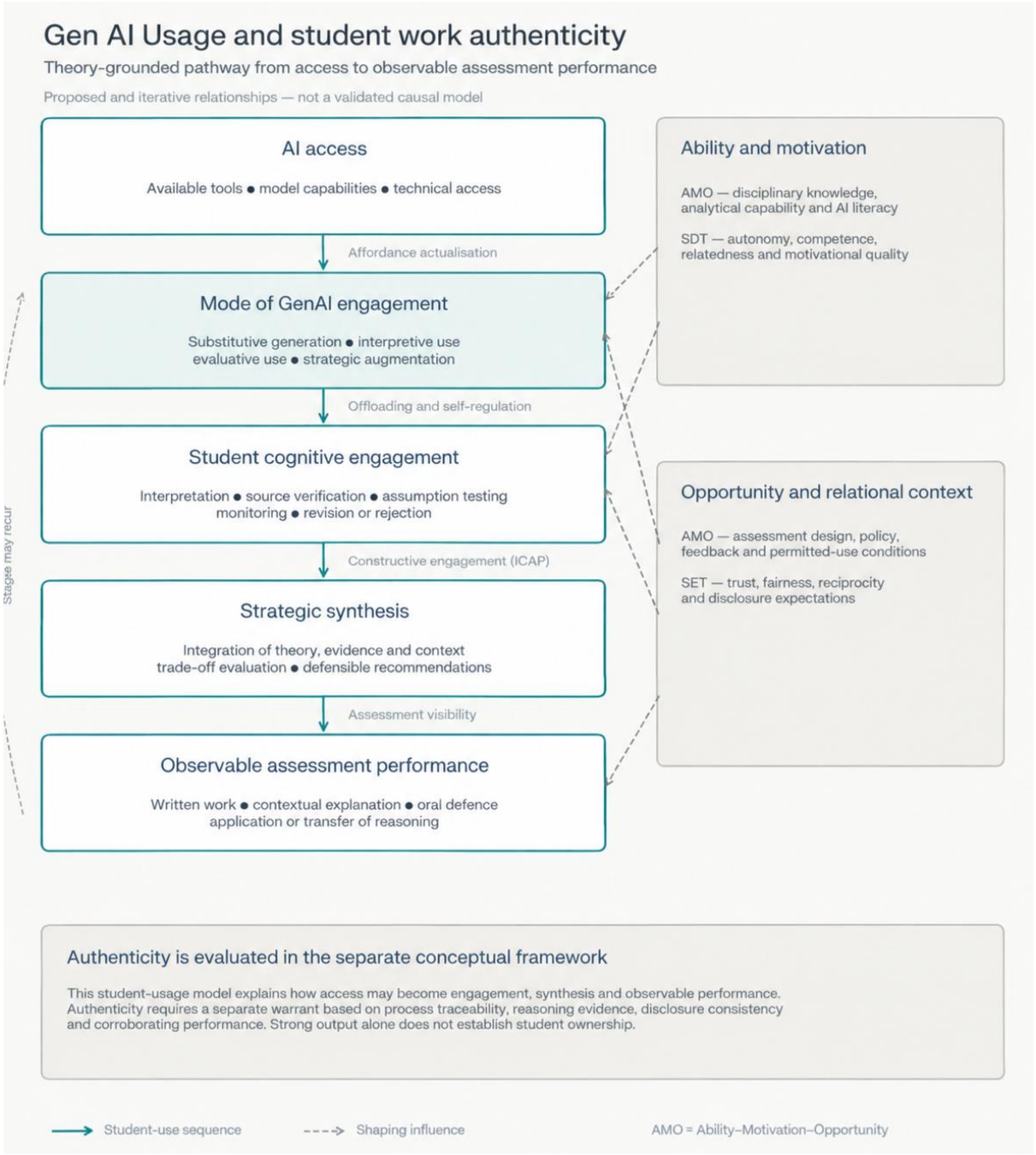


Figure 1: Integrated Conceptual Model explaining the mode of Gen AI engagement amongst students

3.5 Conceptual Framework

Figure 2 explains the conceptual framework where the independent variable is the mode of GenAI engagement. It represents a continuum from substitutive generation, where AI output is accepted with limited scrutiny, through interpretive and evaluative use, to strategic augmentation, where students selectively integrate AI contributions into independently

justified decisions. The model expects more critical and augmentative engagement to produce stronger student reasoning than substitutive use.

The first mediator, cognitive engagement, captures the student's interpretation, source verification, assumption testing, critical evaluation, and revision or rejection of AI output. It explains how a particular mode of GenAI engagement affects the reasoning activities retained by the student.

The second mediator, strategic synthesis, represents the integration of theory, evidence, contextual judgement, competing alternatives, and trade-offs into defensible recommendations. The serial sequence proposes that critical GenAI engagement first supports cognitive engagement, which then enables strategic synthesis.

The dependent variable is **authenticity confidence**, defined as the degree of justified confidence that the reasoning, decisions, and capabilities demonstrated in a submission can be attributed to the student. Authenticity confidence is not equivalent to writing quality, academic performance, disclosure, or the absence of AI. It is an evidence-based attribution judgement.

Authenticity Warrant strength moderates the relationship between strategic synthesis and authenticity confidence. It comprises process traceability, reasoning evidence, disclosure consistency, and corroborating performance. Its role is not to produce student reasoning, but to determine whether observed synthesis can credibly be attributed to the student. Consequently, strategic synthesis should contribute more strongly to authenticity confidence when these evidence sources converge.

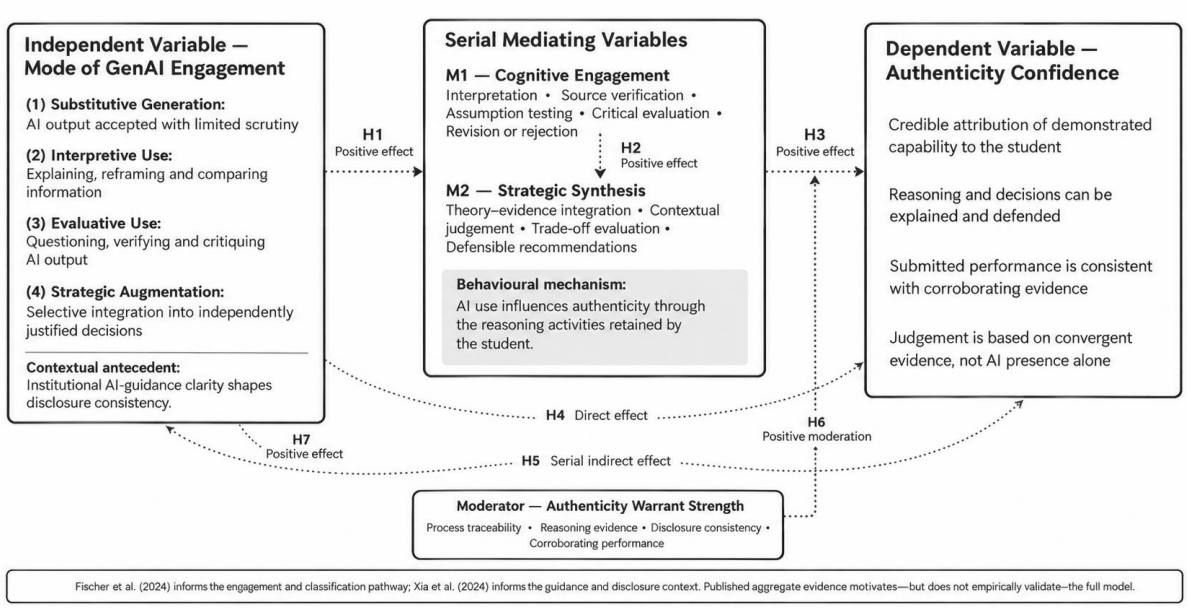


Figure 2: Conceptual framework explaining the model of Gen AI engagement

4. Method

4.1 Research design

This study adopts a **theory-adapting conceptual design supported by structured evidence synthesis and secondary analysis of published aggregate data**. The methodology evaluates whether existing research provides direct, indirect, theoretical, contradictory, or insufficient support for H1–H7. Fischer et al. (2024) provide the principal empirical evidence concerning modes of GenAI engagement, student–assessor classification differences, and descriptive performance patterns. Xia et al. (2024) provide review-level evidence concerning assessment design, institutional policy, responsible use, and disclosure conditions.

The studies are not merged statistically because they use different evidence bases, constructs, samples, and research designs. Instead, their findings are mapped onto specific parts of the conceptual framework. This approach can evaluate the **plausibility and evidential support** of the hypotheses, but it cannot establish causal effects, mediation, or moderation without a common individual-level dataset.

4.2 Evidence base and inclusion criteria

Sources are included when they address at least one model construct or relationship:

Mode of GenAI engagement.

Cognitive engagement.

Strategic synthesis.

Authenticity or capability attribution.

Process traceability and reasoning evidence.

Disclosure consistency.

Corroborating performance.

Institutional AI-guidance clarity.

For each source, the analysis records the research design, context, sample or evidence base, construct definitions, measurements, principal findings, limitations, and relevance to H1–H7. This prevents conceptually different constructs—such as AI-use frequency, cognitive engagement, performance, and authenticity—from being treated as equivalent.

4.3 Operationalisation of constructs

The study evaluates H1–H7 through a structured synthesis of theoretical, empirical, and secondary quantitative evidence. Because the selected studies use different samples, constructs, and research designs, their findings are not combined into a single dataset. Instead, each study is mapped to the specific relationship it can inform within the conceptual framework. Fischer et al. (2024) provide the principal evidence concerning modes of GenAI engagement and differences between student claims and assessor evaluations. Xia et al. (2024) inform the institutional and assessment context, while research on commercial AI detection provides additional evidence concerning the limitations of product-based authenticity judgements.

A hypothesis is not classified as empirically supported unless the relevant constructs and their relationship were measured directly. Where the literature provides only adjacent findings, the hypothesis is described as **partially supported**, **theoretically supported**, or **not directly tested**. This distinction prevents theoretical consistency from being misrepresented as statistical confirmation.

4.4 Data sources and evidence instruments

The analysis uses peer-reviewed studies and published aggregate evidence relevant to the proposed constructs. Fischer et al. (2024) provide the principal engagement-mode classifications and student–assessor comparisons; Xia et al. (2024) inform the institutional and assessment context; additional studies and formal dashboards provide construct-specific evidence for H1–H7. The dashboards are analytic representations of published evidence, not independent participant-level datasets.

4.5 Evidence-collection procedure

Sources were selected when they addressed at least one construct or relationship in the framework. For each source, the analysis recorded the research design, context, sample or evidence base, construct definitions, measurements, principal findings, limitations and relevance to H1–H7. Evidence was then mapped to the specific proposition it could inform, rather than pooled across conceptually different samples and measures.

4.6 Data-analysis strategy

Secondary Analysis

Fischer et al.'s (2024) published aggregate data are used to compare students' claimed engagement with assessor-classified engagement. For each engagement category kkk, the percentage-point difference is calculated as:

$\Delta k = p_{student,k} - p_{assessor,k}$

The overall distributional divergence is calculated using total-variation distance:

$D = \frac{1}{2} \sum_{k=1}^{K} |p_{student,k} - p_{assessor,k}|$

These calculations indicate how far the aggregate student and assessor distributions differ. They do not represent individual misclassification because the published marginal totals do not show how each student was classified by both parties. Published mark differences are interpreted descriptively and not as evidence that a particular GenAI engagement mode caused higher performance.

4.7 Validity, reliability and trustworthiness

Evidence Classification

Evidence for each hypothesis is classified using the following criteria:

Classification	Decision rule
Direct empirical support	The study measures both constructs and tests their relationship
Indirect empirical support	The findings concern closely related constructs but do not test the exact relationship
Theoretical support	Established theory explains why the relationship should occur
Contradictory evidence	Findings are inconsistent with the predicted relationship
Not examined	The study does not measure the relevant constructs or relationship

No numerical evidence score is interpreted as proof. The classification provides a transparent audit trail showing which parts of the framework are empirically informed and which remain original theoretical extensions.

Each hypothesis is assessed using five evidence categories:

Direct empirical support: The source measures both constructs and tests the proposed relationship.

Indirect empirical support: The source examines related constructs or reports findings consistent with the proposed relationship.

Theoretical support: An established theory explains the expected relationship.

Contradictory evidence: Published findings challenge the expected relationship.

Not examined: The source does not measure the constructs required to evaluate the relationship.

The final assessment will be recorded in a hypothesis–evidence matrix:

The methodology does not seek to "prove" H1–H7. It determines whether each hypothesis is directly supported, indirectly supported, theoretically supported, contradicted, or not examined by the available evidence. Actual conclusions regarding each hypothesis should be reported in the Findings or Discussion section.

Hypothesis	Main evidence	Methodological status
H1	Fischer et al.; cognitive-engagement theory	Indirect empirical and theoretical support
H2	ICAP, self-regulation, strategic-reasoning literature	Theoretical and qualitative support
H3	Authentic assessment and attribution literature	Theoretically supported; not directly tested

Hypothesis	Main evidence	Methodological status
H4	Fischer et al.'s classifications and discrepancies	Indirectly supported; direct effect not estimated
H5	Combined evidence relating to H1–H3	Theoretically coherent; mediation not tested
H6	Detector limitations and convergent authenticity evidence	Indirectly supported; moderation not tested
H7	Xia et al.; assessment-policy and SET literature	Review-level and theoretical support

4.8 Ethical considerations

The paper analyses published aggregate evidence and does not use identifiable student-level data. No attempt is made to infer individual misconduct from aggregate classifications or AI-detection indicators. The analysis also avoids presenting proxy measures as direct observations of authenticity confidence, mediation or moderation. Any future participant-level study of the framework should obtain appropriate ethics approval, informed consent and data-protection safeguards, particularly where student work, prompts, drafts or oral-defence records could influence academic decisions.

5. Results

5.1 Evidence-base characteristics

This chapter evaluates H1–H7 through structured evidence synthesis, secondary analysis of Fischer et al.'s (2024) published aggregate results, and comparison with research on GenAI assessment, institutional guidance, and AI-writing detection. The evidence does not permit statistical testing of the complete serial-mediation and moderation model because no single

dataset measures the independent variable, both mediators, the moderator, and the dependent variable for the same students. The findings therefore indicate whether each hypothesis is directly supported, indirectly supported, theoretically supported, or not yet empirically tested.paste.txt+1

The overall assessment is that H1 has the strongest empirical foundation, H2 receives substantial theoretical support, and H3–H6 represent progressively more original extensions beyond the existing evidence. H7 receives contextual support from the assessment and policy literature, although the exact relationship between guidance clarity and disclosure consistency has not been estimated directly.

5.2 Descriptive findings

Evidence Summary

Hypothesis	Assessment	Evidence level
H1: Engagement mode → cognitive engagement	Consistent with available evidence	Indirect empirical and theoretical support
H2: Cognitive engagement → strategic synthesis	Plausible and theoretically supported	Theoretical, empirical and qualitative support
H3: Strategic synthesis → authenticity confidence	Conceptually supported but untested	Original theoretical extension
H4: Engagement mode → authenticity confidence	Plausible but not directly tested	Indirect support
H5: Serial mediation	Coherent but not statistically demonstrated	Integrated theoretical proposition

Hypothesis	Assessment	Evidence level
H6: Authenticity Warrant moderation	Strongly motivated but not tested	Original contribution with indirect empirical support
H7: Guidance clarity → disclosure consistency	Consistent with review evidence	Contextual and theoretical support

5.3 Findings by research question

Across the evidence base, engagement mode appears most informative when it is interpreted through visible reasoning rather than prompt frequency or textual polish. Evidence relevant to RQ1 supports a distinction between low-level generation and more evaluative forms of interaction, although the complete pathway to strategic synthesis remains untested. For RQ2, confidence in authenticity is strengthened when process, reasoning, disclosure and corroborating evidence converge. For RQ3, the reviewed literature supports clearer institutional guidance, but it does not yet quantify the relationship between guidance clarity and disclosure consistency in a matched student sample.

5.4 Proposition results

5.4.1 H1: GenAI engagement and cognitive engagement

H1: More critical and strategically augmentative modes of GenAI engagement are positively associated with cognitive engagement. The evidence is broadly consistent with H1. An **Engagement and Cognition dashboard** (Figure 3) has been created based on Fischer et al.'s (2024) study that distinguishes between relatively generative activity and more interpretive or evaluative engagement. The progression from scanning and generating to interpreting, rephrasing, critiquing, and evaluating corresponds conceptually with increasing student participation in the reasoning process.

Fischer et al. (2024) classified AI-assisted sensemaking into three stages and compared students' claims with assessors' evaluations. Among 118 classified cases, students claimed scanning and generating in 31 cases, interpreting and rephrasing in 67 cases, and critiquing and evaluating in 20 cases. Assessors classified 24, 82, and 12 cases in those categories, respectively.

This interpretation is reinforced by cognitive-offloading and self-regulated-learning theory. GenAI may reduce routine cognitive demands without weakening learning when the student continues to monitor, verify, evaluate, and revise its output. By contrast, offloading becomes substitution when the technology performs the reasoning that the assessment is intended to evaluate and the student accepts the output without substantive scrutiny (Risko & Gilbert, 2016; Zimmerman, 2002).

The ICAP framework provides further support. Interpretive and evaluative activities are more consistent with constructive engagement than merely receiving or reproducing generated information (Chi & Wylie, 2014). However, prompt sophistication and interaction volume cannot establish cognitive engagement independently. A student may instruct GenAI to critique a proposal without personally evaluating the critique.

Fischer et al. 's student–assessor differences strengthen this qualification. Students claimed critiquing and evaluating more frequently than assessors identified it. One interpretation is that critical engagement occurred but remained invisible in the final submission. An alternative is that students interpreted their activity as critical even when the resulting work did not demonstrate substantive evaluation. The aggregate evidence cannot distinguish between these explanations.

Assessment of H1: H1 receives indirect empirical and strong theoretical support, as illustrated in Figure 3, by leveraging Fischer et al's study. The direction of the relationship is plausible,

but the available data do not provide a direct association between engagement mode and an independently measured cognitive-engagement construct.

DESCRIPTIVE EVIDENCE · CONCEPTUAL LINKAGE

H1 evidence and engagement pathway

A distinct evidence panel separating published engagement classifications from the proposed strategic-augmentation extension.

Status: H1 receives indirect descriptive and theoretical support

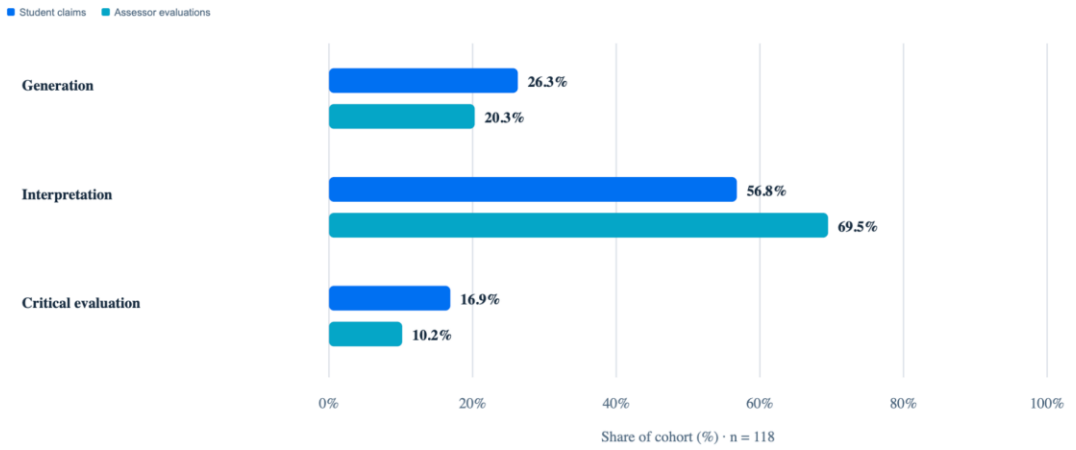
Assessor: interpretation or critical evaluation
79.7%
94 / 118 students · (82 + 12) ÷ 118

Requested KPI · definition pending
79.7%
User-specified value. Construct and denominator not supplied; not a separate measured finding.

Largest marginal classification gap
12.7 percentage points
Interpretation: 69.5% assessor vs 56.8% claims · (82 − 67) ÷ 118

Student claims versus assessor evaluations

Grouped horizontal comparison · highest classified engagement stage



PUBLISHED STAGE	STUDENT CLAIMS	ASSESSOR EVALUATIONS
Scanning and generating	31 · 26.3%	24 · 20.3%
Interpreting and rephrasing	67 · 56.8%	82 · 69.5%
Critiquing and evaluating	20 · 16.9%	12 · 10.2%

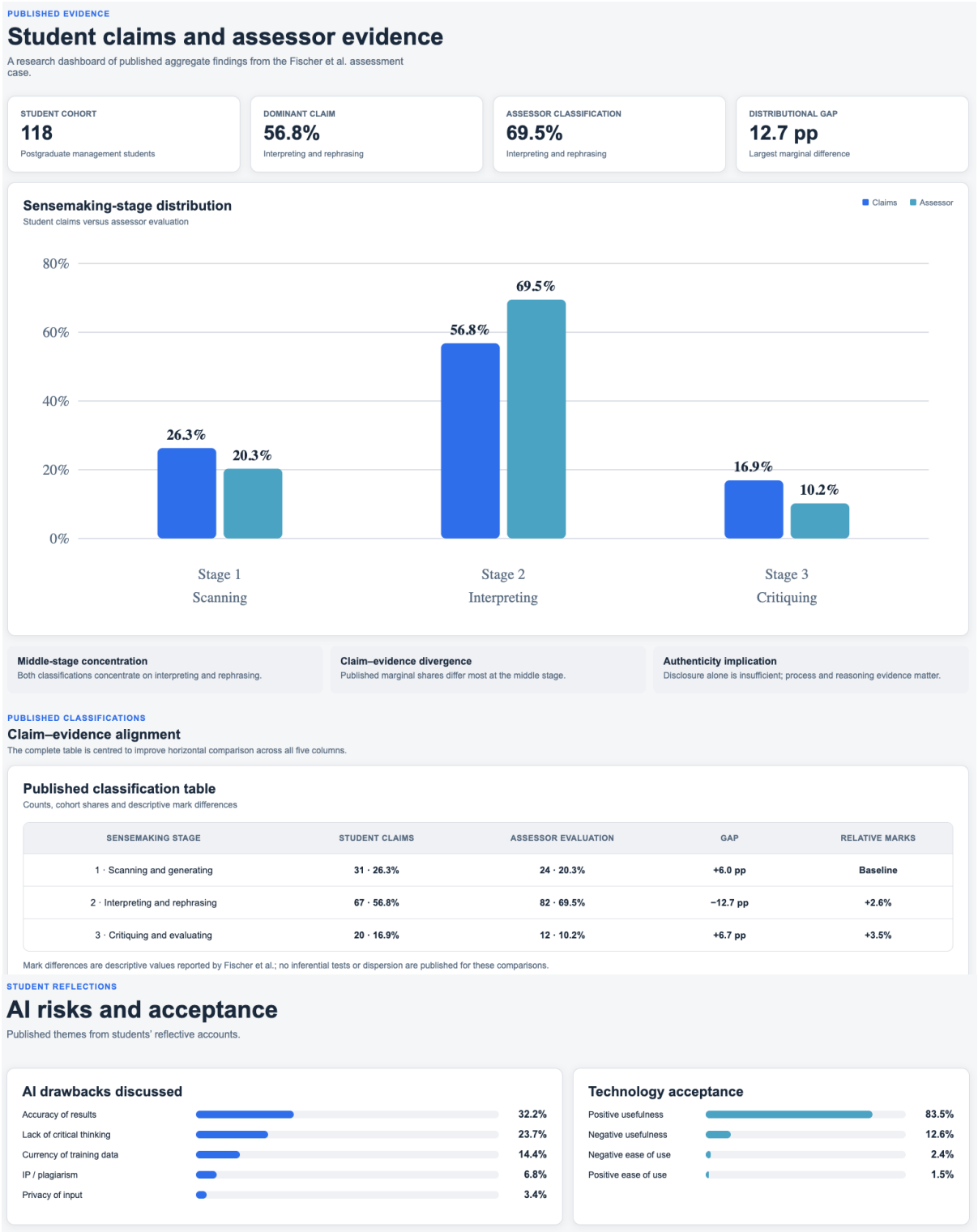


Figure 3: The H1 Engagement and Cognition dashboard visualizes Fischer et al.'s (2024) study of 118 postgraduate management students, showing that students predominantly used GenAI for writing, paraphrasing, and rephrasing, while only around one-tenth demonstrated advanced critical sensemaking.

5.4.2 H2: Cognitive engagement and strategic synthesis

Cognitive engagement → strategic synthesis

H2 proposed that cognitive engagement is positively associated with strategic synthesis. In this study, cognitive engagement refers to the learner's active mental investment in interpreting, questioning, evaluating, and refining information, rather than simply accepting or reproducing an AI-generated response. Strategic synthesis represents the next stage in this process, in which students combine relevant GenAI contributions with their own disciplinary knowledge, evidence, reasoning, and judgement to produce a coherent response. The theoretical expectation underlying H2 is therefore that students who engage more deeply with AI-generated material will be better positioned to select, integrate, adapt, and justify that material within their own work.

The empirical evidence shown in Figure 4, is consistent with this proposed relationship as developed in **H2 Engagement and Cognition Dashboard** below. Lee et al. (2024) conducted a randomized controlled trial involving 61 undergraduate chemistry students, comparing conventional ChatGPT use with a guidance-based ChatGPT learning aid that required students to formulate an initial response and then refine it through hints and iterative feedback. Students in the guided condition recorded higher cognitive engagement than those using standard ChatGPT (34.5 versus 27.4). They also achieved higher post-intervention scores in three capabilities closely related to strategic synthesis: critical thinking (17.3 versus 13.8), problem-solving (15.9 versus 13.1), and creativity (11.0 versus 10.1). The reported effects were statistically significant for critical thinking, $F=41.73$, problem-solving, $F=38.70$, and creativity, $F=8.98$, with partial eta-squared values of 0.418, 0.400, and 0.134 respectively. These findings suggest that an approach designed to sustain students' cognitive participation, rather than

supplying complete answers, is associated with stronger higher-order capabilities required to evaluate, combine, and transform information.

Fischer et al.'s findings offer additional contextual support. Their analysis indicated that many students used GenAI mainly for generating, writing, or rephrasing content, while comparatively few demonstrated critical evaluation. This suggests that GenAI use does not automatically lead to strategic synthesis. Instead, synthesis is more likely when students move beyond reproducing AI output and actively interpret, question, verify, and refine it.

A systematic review of 63 empirical studies similarly found stronger critical-thinking outcomes when GenAI use involved structured prompts, reflection, iterative dialogue, and evaluation. Unguided use, however, increased the risks of cognitive offloading and uncritical acceptance. Therefore, **H2 receives indirect empirical and theoretical support**. Nevertheless, because strategic synthesis was not independently measured, the relationship cannot be regarded as statistically confirmed.

Assessment on H2 Indirect empirical and theoretical support, but it has not been directly tested by the available secondary evidence.

H2: Cognitive engagement is positively associated with strategic synthesis.

> Indirect empirical support — exact H2 relationship not directly tested.

Published aggregate engagement and critical-thinking results provide context for the proposed pathway. Neither numerical panel measures an engagement–strategic-synthesis association.

Reported engagement means

34.5 vs 27.4

Guided versus standard ChatGPT.

Descriptive difference

+7.1 points

Difference in raw means; not an adjusted effect.

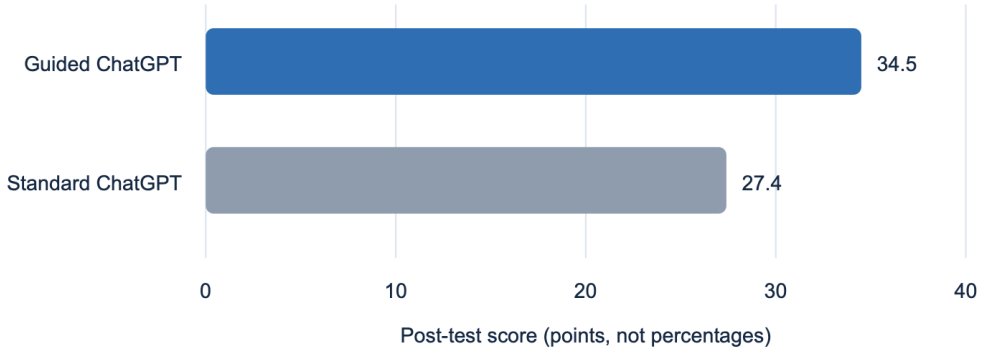
Pooled critical-thinking effect

g = 0.544

95% CI: 0.314–0.775.

Cognitive engagement comparison

Lee et al. (2024): reported post-test means under guided and standard ChatGPT use.



> The 7.1-point difference is descriptive, not a strategic-synthesis score. Error bars are omitted because dispersion values were not verified.

Source: Empowering ChatGPT with guidance mechanism in blended learning: effect of self-regulated learning, higher-order thinking skills, and knowledge construction (2024).



Figure 4: The H2 Engagement and Cognition Dashboard visually evaluates the proposed relationship between students' cognitive engagement with GenAI and their strategic synthesis of AI-supported material. It combines a five-stage reasoning pathway with empirical and literature evidence, while concluding that H2 has indirect empirical and theoretical support but has not yet been directly statistically tested.

5.4.3 H3: Strategic synthesis and authenticity confidence

H3 proposed that strategic synthesis is positively associated with authenticity warrant strength. Strategic synthesis occurs when students critically select, combine, adapt, and contextualize GenAI-generated information using their own knowledge, evidence, reasoning, and judgment. Authenticity warrant strength refers to the quality of evidence supporting the inference that the submitted work reflects the student's intellectual contribution and demonstrated capability.

As explained below in Figure 5, **H3 Evidence Test Dashboard** operationalizes these concepts through two five-dimensional assessments. Strategic synthesis is assessed through selection, critical evaluation, integration, adaptation and justification. Authenticity-warrant strength is

assessed using process traceability, revision coherence, source verification, oral defence and knowledge transfer. Each indicator is rated from 0 to 4 using evidence from drafts, prompts, source checks, reflections and oral defence. The dashboard then calculates the Pearson correlation, significance level and 95% confidence interval from student-level scores

Existing evidence provides indirect support for H3. Fischer et al.'s study of 118 postgraduate students found that GenAI was used mainly for writing, paraphrasing and rephrasing, while only approximately 10% demonstrated advanced sensemaking. However, Fischer et al. did not measure both H3 constructs or test their association.

Under the dashboard's predefined rule, H3 is supported when the relationship is positive, $p<.05p < .05p<.05$, and the 95% confidence interval excludes zero. As no student-level dataset has yet been analyzed, H3 is currently **theoretically and contextually supported but not statistically confirmed**. The dashboard therefore functions as an evidence-organization and hypothesis-testing instrument, not as proof of H3

Assessment on H3 - A Shift from detecting AI-generated language to evaluating student intellectual governance and the evidence supporting attribution. The literature supports each side of the proposed mechanism but does not yet supply a direct student-level test.

HYPOTHESIS 3

Strategic synthesis → authenticity warrant strength

● Indirect support only

A transparent evidence map and direct-test workspace for evaluating whether stronger student synthesis is associated with stronger evidence of intellectual contribution. The dashboard tests evidence; it does not claim mathematical proof.

Current conclusion: H3 is theoretically and contextually plausible, but it remains untested as a direct statistical relationship. Use the "Run direct test" view with coded student-level data to evaluate it.

CASE-STUDY COHORT	ADVANCED SENSEMAKING	DIRECT H3 TESTS FOUND	EVIDENCE STATUS
118	≈10%	0	Contextual
Postgraduate management students	Approximately one-tenth of the cohort	No study in the current evidence set measured both constructs	Useful for rationale, insufficient for confirmation

Hypothesized mechanism

What H3 proposes—not what current evidence has statistically established.

Strategic synthesis
Critical selection, evaluation, adaptation, integration, and justification of AI-supported material.

→ **Positive association**

Authenticity warrant strength
Converging evidence that the work reflects the student's reasoning, capability, and intellectual contribution.

Assessment verdict

Language suitable for the present evidence base.

⚠ **Indirectly supported; not statistically confirmed**

Fischer et al. show that advanced critical sensemaking was uncommon, while assessment-validity literature explains why visible synthesis and defence can strengthen attribution. Neither directly estimates the H3 association.

UPDATED EVIDENCE AUDIT

What the literature can—and cannot—show

Recent studies sharpen the proposed pathway from evaluative AI use to observable authorship, but a direct link between the two operationalized H3 constructs is still missing.

What changed: the evidence base now contains closer qualitative analogues of strategic synthesis—human judgment, corroboration, selective integration, personalization, reconstruction, and abandonment. It still does not contain validated H3 measures or a student-level association estimate.

Observed AI-use pattern

Fischer et al. case study, n = 118.



The authors report that only one-tenth of the cohort reached advanced sensemaking; the ≈90% segment is the derived complement. Fischer et al. (2024)

Evidence ladder

Higher levels provide stronger warrant for H3.

4	**Direct association** Student-level synthesis and warrant scores analysed together.	Missing
3	**Validated measures** Reliable rubrics for both constructs with independent raters.	Missing
2	**Contextual empirical evidence** Observed AI practices, authenticity concerns, verification, reconstruction, and assessor evaluation.	Expanded
1	**Theoretical rationale** Process evidence and oral defence make reasoning more visible.	Available

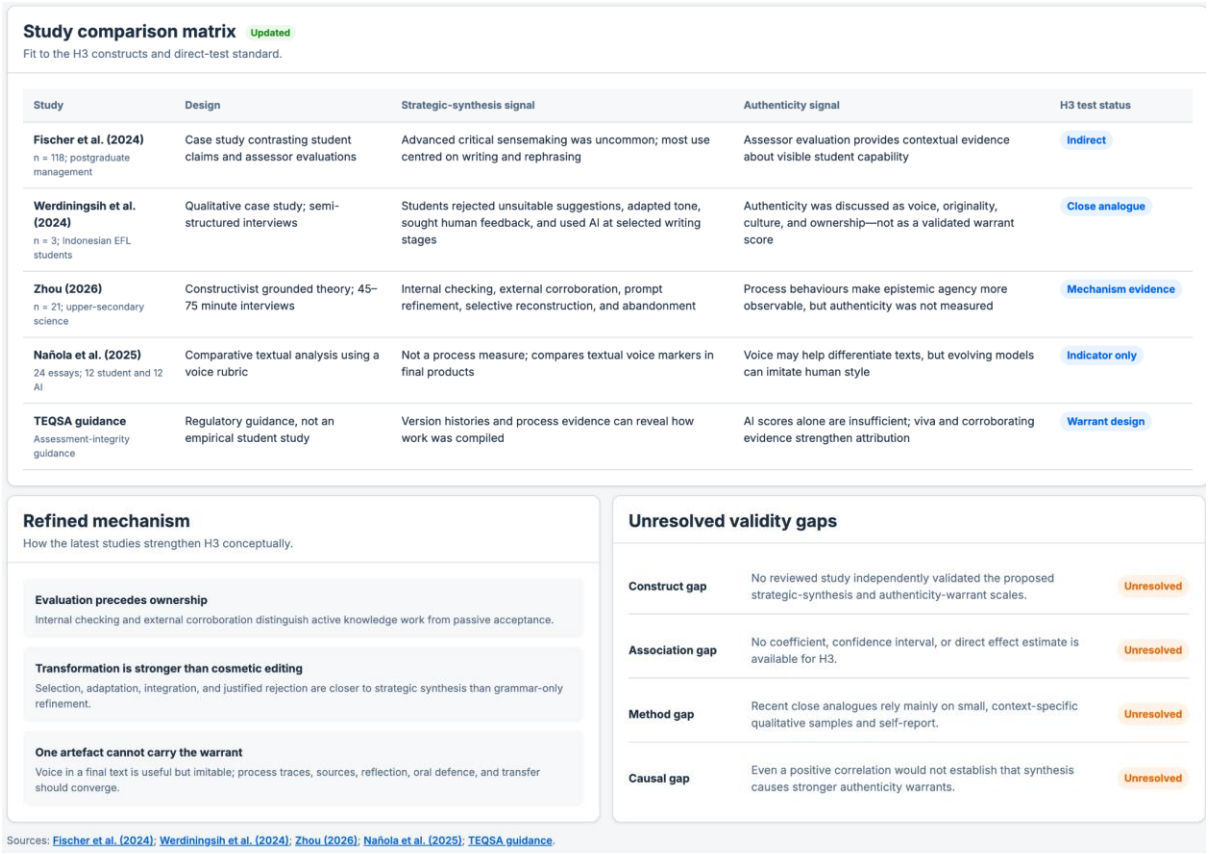


Figure 5: H3 Evidence Test Dashboard is an interactive framework for examining whether students' strategic synthesis of GenAI content is associated with stronger evidence of authentic intellectual contribution. It combines a two-construct assessment rubric, evidence review and statistical testing—including Pearson's rrr, ppp-value and 95% confidence interval—while distinguishing conceptual support from direct empirical confirmation

5.4.4 H4: Engagement mode and authenticity confidence

Critical and strategically augmentative engagement with generative AI is positively associated with assessor confidence that submitted work authentically represents the student's knowledge, reasoning and contribution, whereas substitutive engagement is negatively associated with assessor authenticity confidence.

H4 treats authenticity as an evidential judgment rather than a binary determination of whether AI was used. The central proposition is that assessor's confidence depends on how GenAI is incorporated into the student's work and whether the resulting process leaves credible evidence of intellectual ownership. Critical engagement involves challenging, checking and correcting AI outputs; strategic augmentation involves using AI selectively while retaining control of the

reasoning and final decisions; and substitutive engagement involves delegating substantive reasoning, judgment or authorship to the system.

The **Authenticity Confidence dashboard** shown in Figure 6, represents H4 as a three-stage evidence chain:

- **Engagement mode:** critical evaluation, strategic augmentation or substitutive delegation.
- **Authenticity warrant:** observable evidence such as student reasoning, verification, justified revisions, disclosure consistency and process traces.
- **Assessor's confidence:** the assessor's judgment that the submitted work validly represents the student's own capability and contribution.

This framing is consistent with the wider shift from assessing only a finished product toward evaluating the process through which it was produced. UNESCO's assessment analysis argues that, when AI can generate polished outputs, development records, iterative reasoning, reflection and dialogue become more informative indicators of learning and intellectual ownership. The mechanism implied by H4 is therefore not that critical AI use automatically proves authenticity; rather, critical or augmentative engagement should produce stronger authenticity warrants, which should increase assessor confidence.

The Authenticity dashboard provides indirect empirical support for this proposition. Oliveira et al. analysed 1,450 GenAI interactions associated with 70 graded essays in STEM writing-intensive courses. Interaction quality was positively associated with essay performance ($r = 0.54$, 95% CI [0.34, 0.68], $p < .001$). Higher-performing patterns reflected collaborative intellectual partnership, purposeful steering, visible expertise and targeted improvement. Weaker patterns involved basic information retrieval or passive task delegation, where students exercised less control over the intellectual process. These findings suggest that how students engage with GenAI is more informative than whether they use it.

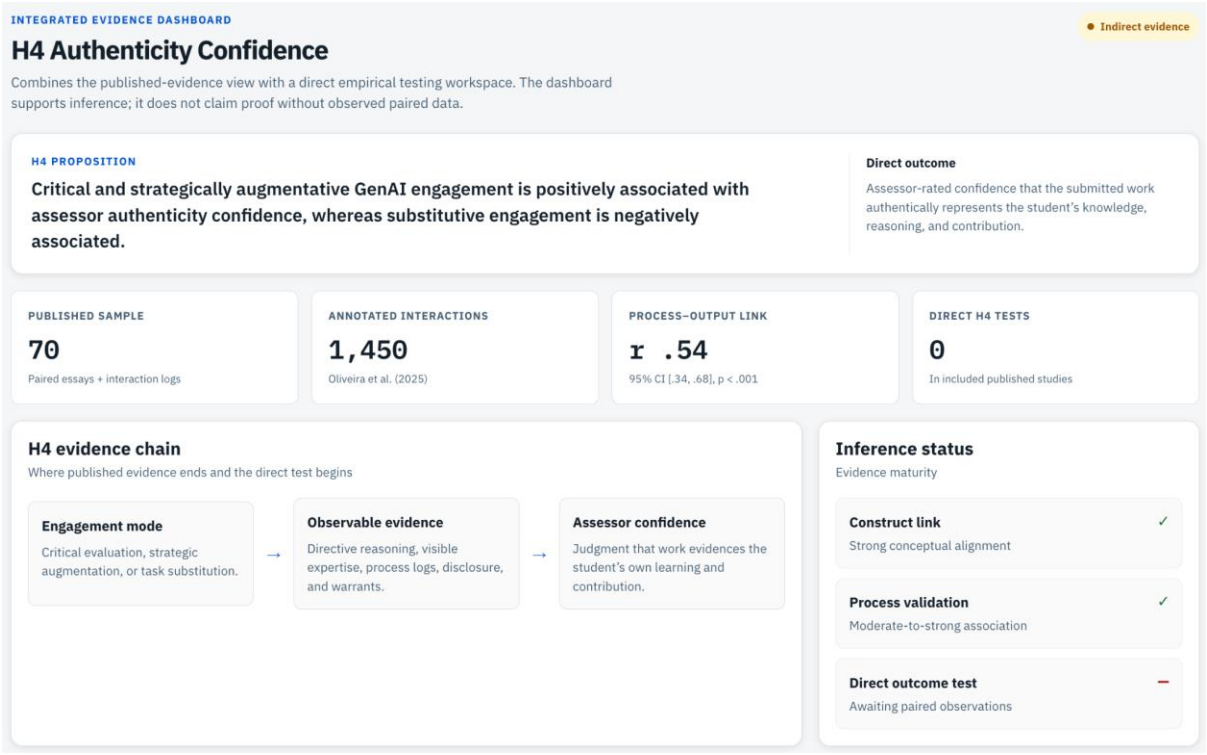
Supporting evidence strengthens this interpretation. Fischer et al. found that postgraduate students used GenAI mainly for writing and paraphrasing, while only about one-tenth demonstrated advanced sensemaking. Nie et al. reported positive associations between GenAI assessment literacy and academic-integrity intention ($r = 0.51$), responsible-use intention ($r = 0.59$), and feedback engagement ($r = 0.60$). UNESCO also recommends shifting assessment attention from finished outputs toward visible learning processes, iteration, critique and reflection.

The central mechanism is authenticity-warrant strength. Critical and augmentative engagement should create observable indicators of student agency, including verification, justified revisions, disclosure consistency, reasoning traces and the ability to defend decisions. When these indicators converge, assessors have a stronger basis for attributing the work to the student's own capability. Substitutive engagement should weaken that basis because the relationship between the product and demonstrated student competence becomes less visible.

Nevertheless, Oliveira et al. measured interaction quality and essay performance, not assessor authenticity confidence. The dashboard therefore contains proxy evidence rather than a direct H4 test. H4 should be regarded as provisionally supported through theoretical logic, construct alignment and converging evidence, but not empirically confirmed. Confirmation requires paired student–assessor data showing positive coefficients for critical engagement and strategic augmentation, a negative coefficient for substitution, and preferably significant indirect effects through authenticity-warrant strength

H4 concludes that the way students engage with GenAI influences assessor confidence in the authenticity of their work. Critical and strategically augmentative engagement should strengthen confidence by demonstrating active judgement, intellectual ownership and purposeful integration of AI-generated material. Conversely, substitutive engagement may reduce confidence because it obscures the student's contribution and weakens evidence of

independent learning. Existing studies provide indirect empirical support for this relationship, but direct confirmation requires paired measures of engagement mode and assessor authenticity confidence. H4 is therefore provisionally supported, subject to confirmatory testing



PUBLISHED EVIDENCE

Evidence gap visible

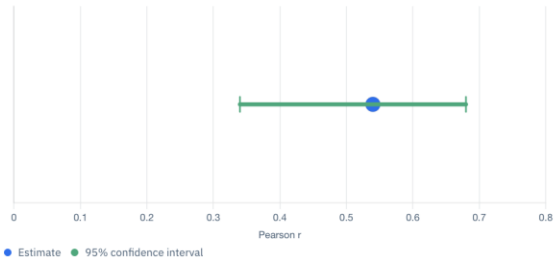
Triangulation, not substitution

The combined evidence supports the plausibility and measurement logic of H4, but no included study directly estimates engagement mode against assessor authenticity confidence.

Oliveira process validation

Association between GenAI interaction quality and essay score

N = 70



Engagement profiles

Observed patterns mapped to H4

Collaborative intellectual partnership +
Idea development, conceptual clarity, critical content.

Targeted improvement partnership +
Text refinement with purposeful steering.

Basic information retrieval −
Shallow retrieval with limited knowledge infusion.

Passive task delegation −
AI executes the task with limited student control.

Evidence register

What each source can—and cannot—establish for H4

STUDY	DESIGN / SAMPLE	FINDING	H4 CONTRIBUTION	DIRECTNESS
Oliveira source-bundle reanalysis	445 student-course records; 103 AI users; 1,450 interactions	AI-interaction score correlated with essay score: Pearson r = .434; Spearman ρ = .471; p < .001.	Provides reproducible process/performance proxy evidence with cohort filtering.	Proxy
Oliveira et al. (2025)	70 essays; 1,450 prompts	Interaction quality correlated with essay quality: r = .54, 95% CI [.34, .68].	Validates process traces and differentiates agentic from delegative profiles.	Indirect
Fischer et al. (2024)	118 postgraduate students	Most use focused on writing/paraphrasing; about one-tenth reached advanced sensemaking.	Supports the distinction between surface/substitutive and critical engagement.	Contextual
Kofinas et al. (2025)	Experimental mixed evidence	Authentic tasks alone may not secure integrity; process evidence remains important.	Explains why confidence should draw on converging process and product evidence.	Boundary
Nie et al. (2026)	486 + 798 students	GenAI assessment literacy correlated with integrity intention (r = .51) and responsible-use intention (r = .59).	Triangulates critical verification, ethical attribution, and responsible revision.	Indirect

Inference boundary: Oliveira et al. measured process quality and essay performance—not assessor authenticity confidence. The direct H4 test therefore requires the paired design implemented in the next workspace.

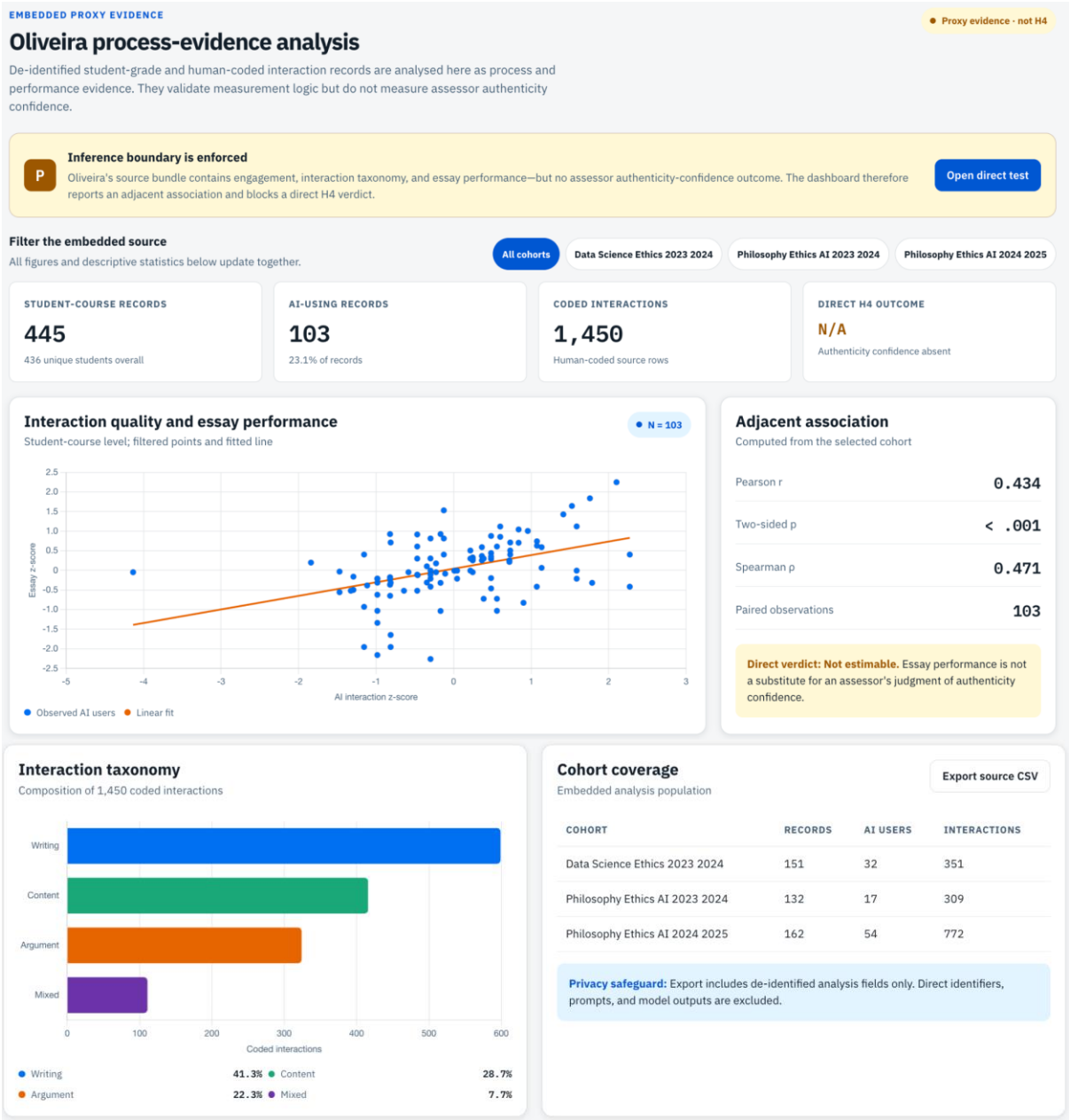


Figure 6: The H4 Authenticity Confidence Dashboard translates Oliveira et al.'s (2024) work into an interactive evidence base for examining how students engage with GenAI. It incorporates 445 student-course records, 436 students and 1,450 human-coded interactions across three cohorts, enabling analysis of interaction quality, engagement profiles and essay performance.

5.4.5 H5: Serial mediation

H5 proposes that cognitive engagement, strategic synthesis and authenticity-warrant strength serially mediate the relationship between students' GenAI engagement mode and assessor authenticity confidence. More specifically, critical and strategically augmentative engagement with GenAI is expected to stimulate active questioning, verification

and reflection. This cognitive engagement should enable students to select, integrate and transform AI-generated material strategically rather than reproduce it passively. Strategic synthesis should then produce stronger authenticity warrants—such as documented prompts, source verification, drafts, revisions and reflective explanations—which increase an assessor's confidence that the submitted work represents the student's knowledge, judgement and intellectual contribution. As shown in Figure, **Serial Mediation Evidence Dashboard** visualizes this proposed mechanism as a five-stage pathway:

GenAI engagement mode → cognitive engagement → strategic synthesis → authenticity-warrant strength → assessor authenticity confidence.

Its evidence map and construct-coverage matrix distinguish direct measures from proxy, contextual and theoretical evidence. This distinction is important because the dashboard builds a cumulative case for the plausibility of H5 but does not treat evidence from separate studies as confirmation of the complete serial pathway. The dashboard therefore makes the inferential boundary explicit: a single participant-level dataset containing all five constructs is required to test H5 directly.

Oliveira et al. (2025) provides the strongest empirical support for the early stages of the mechanism. Their **DRIVE framework** conceptualizes student–GenAI interactions through **Directive Reasoning Interaction**, reflecting how students steer the interaction, and **Visible Expertise**, reflecting how they articulate domain knowledge within that interaction. The study examined 1,450 annotated interactions from 70 graded essays and reported a positive association between interaction quality and essay performance,

r=0.54r=.54r=0.54, p<.001p<.001p<.001.

The dashboard interprets these measures cautiously as proxies for cognitive engagement and strategic synthesis: collaborative intellectual partnership and targeted improvement indicate

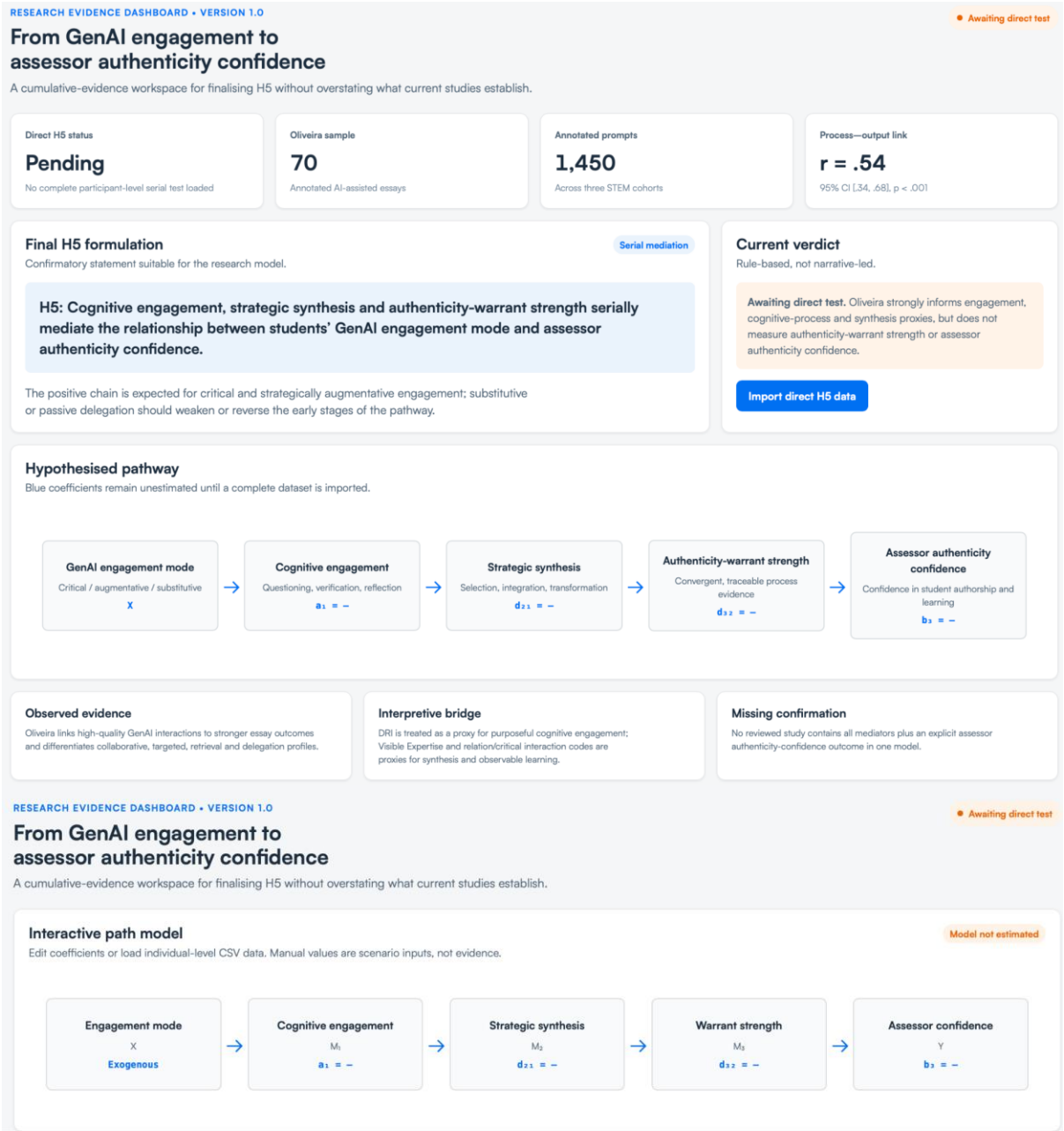
purposeful engagement, whereas basic information retrieval and passive task delegation suggest weaker cognitive involvement.

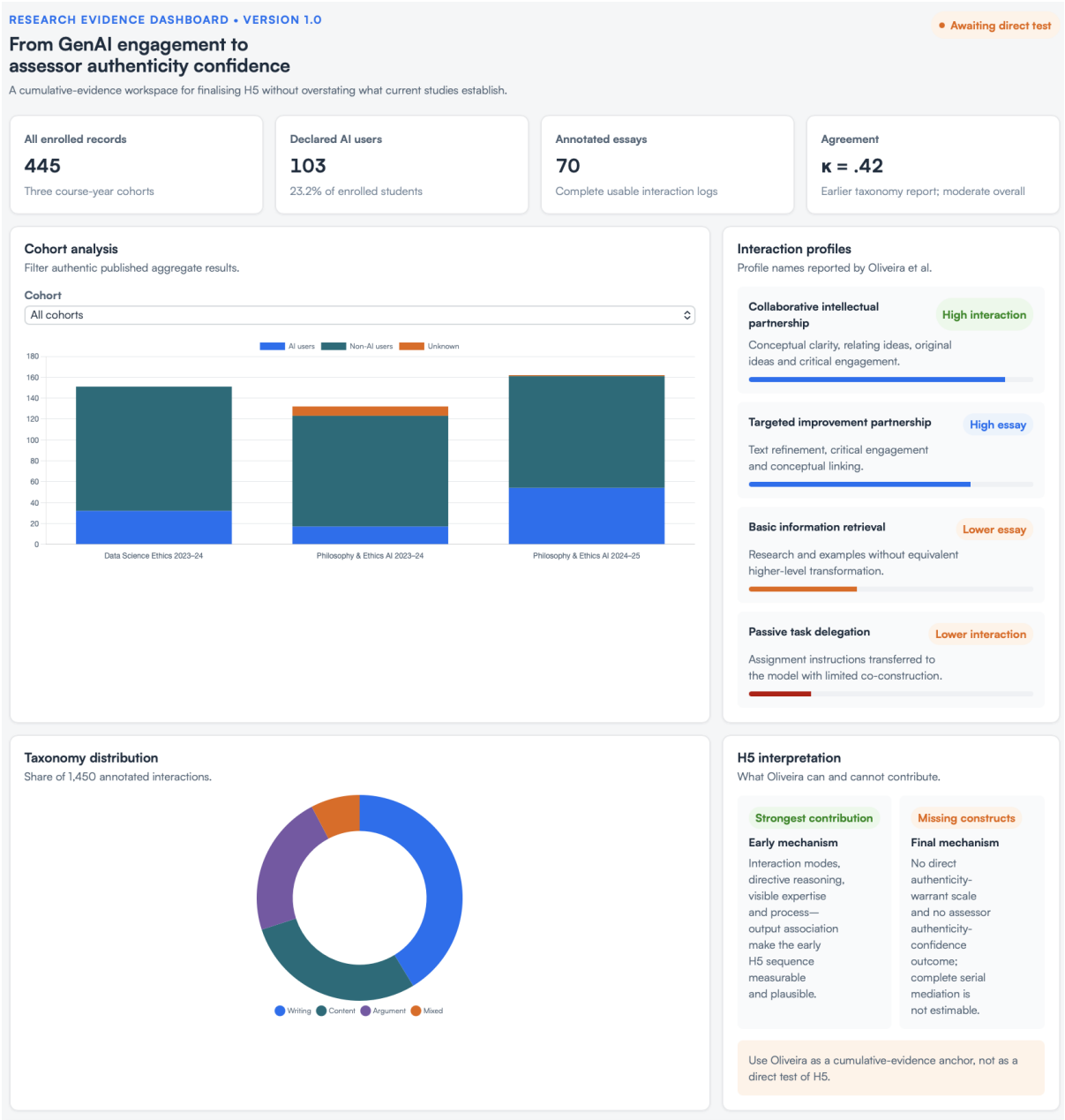
This interpretation is reinforced by Yang et al. (2025), who analysed 1,445 GenAI-assisted writing sessions. Students who actively modified GenAI-generated text produced improvements in lexical sophistication, syntactic complexity and textual cohesion, whereas frequent acceptance of unmodified AI output was associated with lower writing quality. However, neither Oliveira et al. nor Yang et al. directly measured authenticity-warrant strength or assessor confidence. Fleckenstein et al. (2024) further demonstrated why these later stages matter: novice and experienced teachers struggled to distinguish AI-generated from student-written essays and were overconfident in their judgments. This supports the need for convergent process evidence rather than assessor intuition alone, but it remains contextual evidence rather than a direct test of the final H5 path

Accordingly, the dashboard classifies H5 as **awaiting direct test**, rather than supported or rejected. Confirmatory analysis should estimate the complete indirect effect a1×d21×d32×b3, alongside all shorter specific indirect pathways, using at least 5,000 bootstrap resamples. H5 would be supported if the confidence interval for the complete serial indirect effect excludes zero and the constituent paths follow the hypothesised direction. If only shorter indirect pathways are significant, H5 should be classified as partially supported. Current evidence therefore provides a strong theoretical and empirical foundation for H5 while appropriately reserving statistical confirmation for the proposed participant-level study.

H5 concludes that GenAI use influences perceptions of student-work authenticity through a sequential process rather than a direct relationship. Critical and augmentative engagement promotes deeper cognitive engagement, enabling students to synthesize AI-generated material strategically and produce stronger evidence of their intellectual contribution. These authenticity warrants should subsequently increase assessor confidence in the work's authenticity.

Although existing studies support individual stages of this mechanism, the complete serial pathway remains untested. H5 is therefore theoretically and empirically plausible but requires confirmatory participant-level analysis before it can be accepted.





Published cohort data

Only aggregate counts are embedded; no confidential student data.

Cohort	AI users	Non-AI	Unknown	Total	Annotated essays	Annotations
Data Science Ethics 2023—24 (BSc)	32	119	0	151	21	369
Philosophy & Ethics AI 2023—24 (MSc)	17	106	9	132	16	309
Philosophy & Ethics AI 2024—25 (MSc)	54	107	1	162	33	772

Figure 7: The H5 Serial Mediation Evidence Dashboard visualizes how GenAI engagement may influence assessor authenticity confidence through cognitive engagement, strategic synthesis and authenticity-warrant strength

5.4.6 H6: Authenticity Warrant moderation

H6 proposes that evidence of strategic synthesis does not automatically give assessors confidence that an AI-assisted submission represents the student's own capability. Strategic synthesis refers to the student's ability to evaluate, integrate, challenge and transform AI-generated content into a coherent response. Assessors also require corroborating evidence connecting this reasoning to the student. Such evidence may include interaction histories, progressive drafts, source trails, reflective explanations and oral defense. Collectively, these elements form an authenticity warrant that strengthens the attribution of demonstrated capability to the student.

As explained in Figure 8, **Authenticity Warrant Moderation Dashboard** examined this proposition using Oliveira et al.'s **DRIVE study**. The DRIVE framework evaluates learning through *Directive Reasoning*, *Interaction*, and *Visible Expertise*, making students' reasoning, disciplinary knowledge, and control of GenAI interactions observable. Oliveira et al. analyzed 1,450 annotated interactions linked to 70 graded essays and found a significant positive relationship between GenAI-interaction quality and essay performance, $r=0.54$, $p<.00$. This finding indicates that interaction traces provide meaningful process evidence beyond the quality of the final submission.

In this dashboard, strategic synthesis was represented by synthesis-related interaction behaviors, while authenticity-warrant strength was approximated through the volume, breadth, and cross-domain coverage of the interaction evidence. The moderation coefficient was positive for both DRIVE interaction quality, $\beta=.156$, 95% CI [−.069,.382], $p=.175$, and essay performance, $\beta=.156$, 95% CI [−.027,.340], $p=.095$. However, neither interaction reached the conventional 5% significance threshold. The conditional pattern nevertheless showed that the strategic-synthesis slope increased from −.024 at low warrant strength to .289 at high warrant strength.

Accordingly, H6 is **directionally supported through proxy evidence but not empirically confirmed**. The findings are consistent with the proposed moderation mechanism, but Oliveira et al.'s dataset does not contain assessor authenticity-confidence ratings. Direct confirmation therefore requires paired measures of strategic synthesis, authenticity-warrant strength and assessor confidence for the same submissions.

H6 concludes that authenticity-warrant strength is an important boundary condition in translating strategic synthesis into assessor authenticity confidence. The dashboard showed positive interaction coefficients for both DRIVE interaction quality and essay performance, while the conditional slope increased as warrant strength moved from low to high. This pattern is consistent with the proposition that assessors are more likely to attribute sophisticated AI-assisted reasoning to the student when it is supported by credible process evidence, including interaction histories, drafts, reflective explanations and oral defence. However, the interaction effects were not statistically significant, and Oliveira et al.'s dataset did not include a direct measure of assessor authenticity confidence. H6 therefore receives directional proxy support but cannot yet be accepted as empirically confirmed. Confirmation requires paired observations of strategic synthesis, authenticity-warrant strength and assessor confidence for the same submissions, preferably analysed using multilevel moderation where ratings are nested within assessors. H6 therefore remains provisional.

MODERATION HYPOTHESIS · H6

Authenticity-warrant moderation

H6: Authenticity-warrant strength positively moderates the relationship between strategic synthesis and assessor authenticity confidence, such that the relationship is stronger when warrant strength is high.

Primary empirical base: Oliveira et al. (2025), DRIVE framework · public OSF data and reproducible scripts.

CURRENT EVIDENCE VERDICT

Directional proxy support—not confirmed

The interaction estimates are positive, but neither proxy model reaches the conventional 5% significance threshold. The required assessor-confidence outcome is absent.

• DIRECT TEST OUTSTANDING

Enrolled students	AI users	Analysed essays	Prompt annotations	Process—essay correlation
445	103	70	1,450	r = .54
Three STEM courses	23.2% opted in	Usable paired logs	35 taxonomy classes	95% CI [.34, .68]

Hypothesis logic

DRIVE makes student reasoning and expertise observable. H6 extends that logic by asking whether stronger corroborating process evidence changes how assessors attribute synthesis to the student.

Moderated relationship

Conceptual model—not a claim that Oliveira et al. measured all H6 constructs.



Moderation claim: the synthesis—confidence slope should become more positive as warrant strength increases.

What DRIVE contributes

Two observable process dimensions form the empirical bridge.

Directive Reasoning Interaction (DRI)
How actively and purposefully a student steers, critiques and redirects GenAI.

Visible Expertise (VE)
How clearly course knowledge, argumentation and student judgement appear in the dialogue.

Authenticity-warrant proxy
Volume, breadth and cross-domain coverage of the submitted process trace. This is a researcher-derived proxy—not Oliveira et al.'s validated H6 construct.

Hypothesis logic

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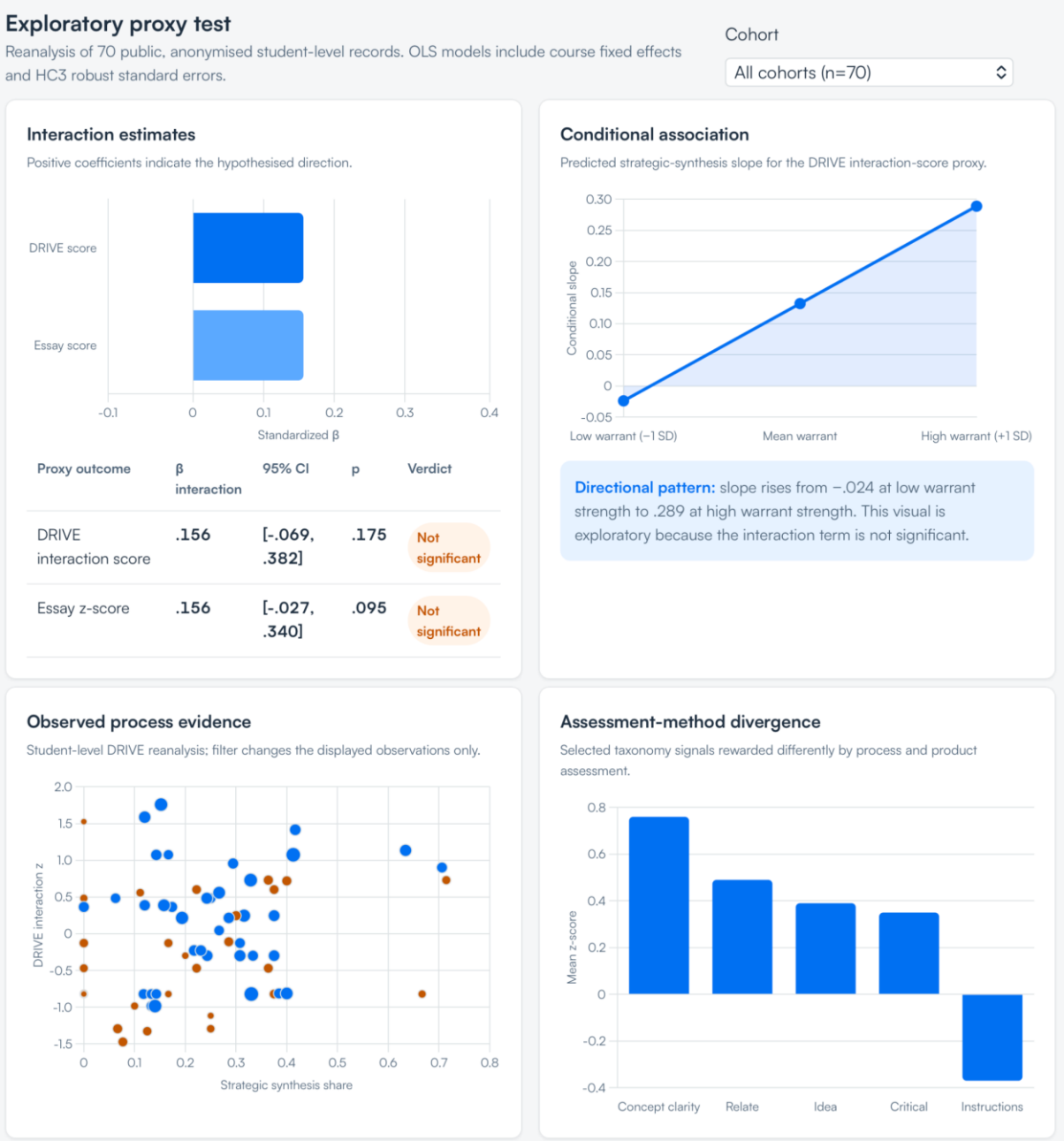


Figure: 8 H6 Authenticity Warrant Moderation Dashboard uses Oliveira et al.'s DRIVE data to visualise how the strength of process evidence may influence the relationship between strategic synthesis and authenticity-related outcomes. Its results provide directional proxy support for H6 while clearly indicating that direct confirmation requires assessor authenticity-confidence data.

5.4.7 H7: Guidance clarity and disclosure consistency

H7 proposes that institutional AI-guidance clarity is positively associated with students' disclosure consistency. The underlying argument is that students are more likely to disclose their use of generative AI consistently when institutions clearly specify what forms of AI assistance are permitted, which activities must be declared, what information should be

included, and how the disclosure should be presented. Conversely, vague or inconsistent guidance may leave students uncertain about whether activities such as brainstorming, editing, summarising, or text generation require acknowledgement.

As presented in the Figure 9, **Guidance Clarity Dashboard**, Xia et al. (2024) retrieved 969 records and included 32 empirical studies in their scoping review. The dashboard shows that 26 of the 32 studies identified institution-level implications, indicating that GenAI had disrupted established assessment practices and affected the formulation of institutional policies. Xia et al. concluded that universities should reconsider their assessment policies and establish clear rules governing GenAI use. They further observed that using GenAI may not necessarily constitute plagiarism when students are transparent about how it has been used in their submissions.

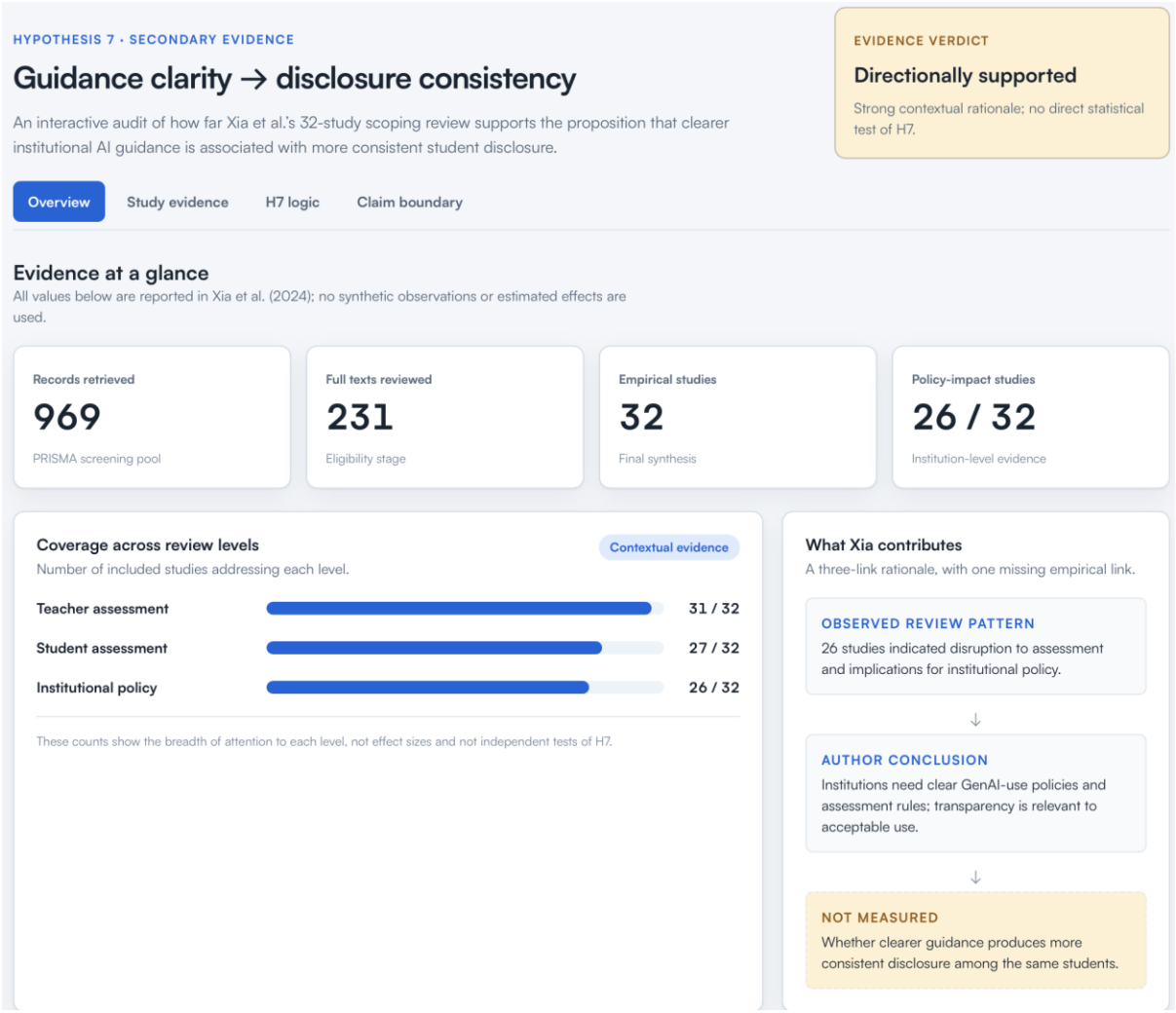
The dashboard translates these findings into the proposed H7 mechanism. Clear guidance reduces ambiguity by defining acceptable use and establishing a shared disclosure standard across assessment contexts. It can also make disclosure psychologically safer by reassuring students that appropriately declared AI assistance will not automatically be treated as misconduct. Guidance clarity may therefore encourage students to report AI use more completely and apply the same disclosure practices across different modules, assessments, and stages of their work. This interpretation is consistent with Xia et al.'s emphasis on clear assessment rules, transparency, responsible learning, and academic integrity

However, the dashboard also identifies an important evidential boundary. Xia et al. did not measure students' perceptions of guidance clarity, assess disclosure consistency, or estimate a statistical relationship between the two constructs. Their study therefore does not directly prove H7. Instead, it provides strong contextual and theoretical support for the predicted relationship by demonstrating the institutional need for clear AI policies and positioning transparency as an essential condition of responsible GenAI use.

On this basis, **H7 is assessed as directionally supported but not empirically confirmed**. The dashboard strengthens the rationale for the hypothesis by combining the review's institutional evidence with its conclusions concerning policy clarity and transparency. A direct empirical test would nevertheless require matched student-level measures of perceived guidance clarity and disclosure consistency, analysed using correlation, regression, or structural equation modelling.

The evidence presented above in H7 Guidance Clarity Dashboard provides **directional support** for the proposition that institutional AI-guidance clarity is positively associated with disclosure consistency. Xia et al.'s (2024) review of 32 empirical studies demonstrates that GenAI has created substantial pressure for institutions to reconsider their assessment policies and establish clearer expectations concerning responsible use, transparency, and academic integrity. Clear guidance should reduce students' uncertainty about when disclosure is required, what information must be reported, and how AI assistance should be acknowledged, thereby encouraging more consistent disclosure practices.

However, Xia et al. did not directly measure guidance clarity and disclosure consistency within the same student sample or estimate the statistical relationship between these constructs. Consequently, the dashboard does not empirically confirm causality or an effect size. **H7 is therefore theoretically and directionally supported, but remains open to direct empirical validation using matched student-level data.**


HYPOTHESIS 7 · SECONDARY EVIDENCE
Guidance clarity → disclosure consistency
An interactive audit of how far Xia et al.'s 32-study scoping review supports the proposition that clearer institutional AI guidance is associated with more consistent student disclosure.
EVIDENCE VERDICT
Directionally supported
Strong contextual rationale; no direct statistical test of H7.
Overview
Study evidence
H7 logic
Claim boundary
Evidence at a glance
All values below are reported in Xia et al. (2024); no synthetic observations or estimated effects are used.
Records retrieved
969
PRISMA screening pool
Full texts reviewed
231
Eligibility stage
Empirical studies
32
Final synthesis
Policy-impact studies
26 / 32
Institution-level evidence
Coverage across review levels
Contextual evidence
Number of included studies addressing each level.
Teacher assessment
31 / 32
Student assessment
27 / 32
Institutional policy
26 / 32
These counts show the breadth of attention to each level, not effect sizes and not independent tests of H7.
What Xia contributes
A three-link rationale, with one missing empirical link.
OBSERVED REVIEW PATTERN
26 studies indicated disruption to assessment and implications for institutional policy.
AUTHOR CONCLUSION
Institutions need clear GenAI-use policies and assessment rules; transparency is relevant to acceptable use.
NOT MEASURED
Whether clearer guidance produces more consistent disclosure among the same students.

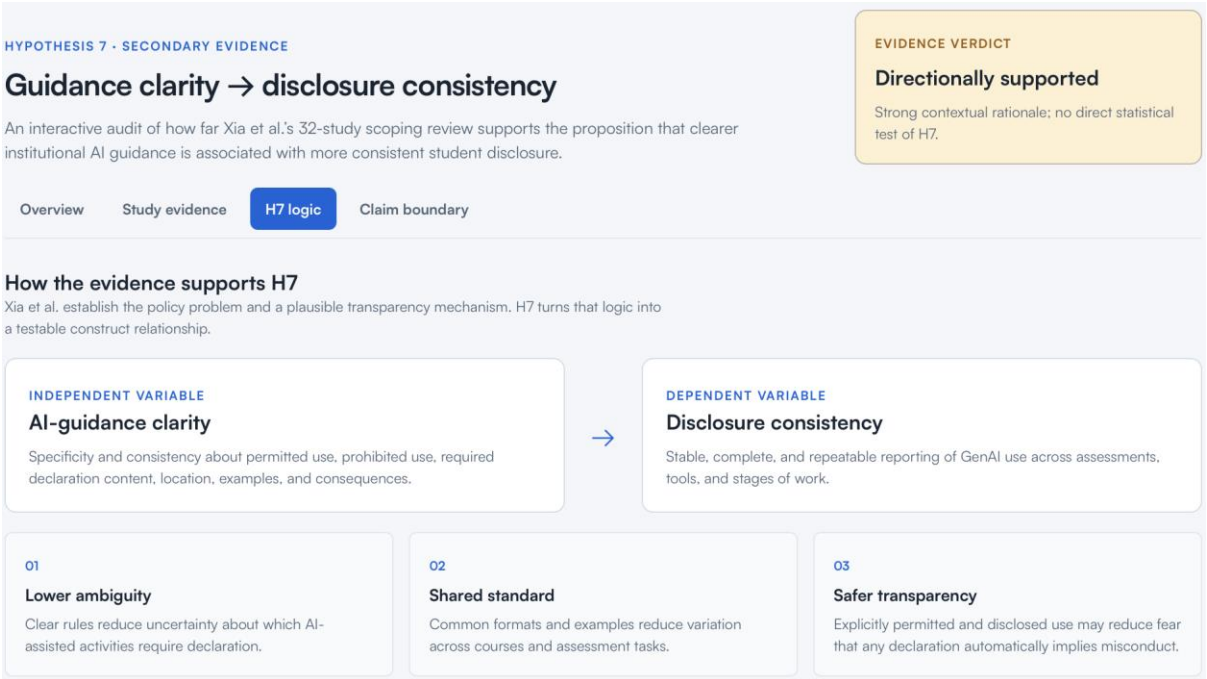


Figure: 9 H7 Guidance Clarity Dashboard synthesises Xia et al.'s (2024) 32-study review to show how the need for clearer institutional AI policies and transparent GenAI use provides directional support for the proposed relationship between guidance clarity and disclosure consistency. However, because the review did not measure both constructs or statistically test their relationship, the dashboard supports H7 theoretically rather than empirically confirming it.

5.5 Integrated authenticity-evidence model

Summary of the Hypotheses

Taken together, the findings support the study's central argument that the authenticity of GenAI-assisted work depends less on whether AI was used and more on how students engaged with it, how visibly they demonstrated their reasoning, and whether institutional conditions encouraged transparent use. Oliveira et al.'s finding that interaction quality was positively associated with essay performance reinforces the importance of evaluating the learning process alongside the final submission.

Hypothesis	Proposed relationship	Main finding	Assessment
H1	Critical and strategically augmentative GenAI engagement is positively associated with cognitive engagement.	Fischer et al.'s engagement patterns indicate that interpretive and evaluative uses involve more active reasoning than scanning or substitutive generation. However, cognitive engagement was not measured as a separate variable.	**Indirect descriptive and theoretical support**
H2	Cognitive engagement is positively associated with strategic synthesis.	Evidence indicates that interpretation, verification and critical evaluation support students' ability to integrate and justify AI-generated material. The precise relationship between the two constructs was not directly tested.	**Indirect empirical and theoretical support**
H3	Strategic synthesis is positively associated with authenticity-warrant strength.	Strategic integration, transformation and justification make students' intellectual contribution more visible. Nevertheless, no available dataset independently measured both constructs for the same students.	**Conceptually and contextually supported**

Hypothesis	Proposed relationship	Main finding	Assessment
H4	Critical and strategically augmentative GenAI engagement is positively associated with assessor authenticity confidence.	Oliveira et al.'s DRIVE evidence shows that interaction quality is positively related to essay performance, with r=.54r=.54r=.54, p<.001p<.001p<.001. This supports the value of observable engagement processes, although assessor authenticity confidence was not measured directly 4tu+1.	**Provisionally supported through indirect empirical evidence**
H5	Cognitive engagement, strategic synthesis and authenticity-warrant strength serially mediate the relationship between GenAI engagement mode and assessor authenticity confidence.	Existing evidence supports the plausibility of individual stages, but no single participant-level dataset measures and tests the complete serial pathway.	**Theoretically plausible; awaiting direct test**
H6	Authenticity-warrant strength positively moderates the relationship between strategic synthesis and assessor authenticity confidence.	The dashboard produced positive directional proxy patterns, but the interaction effects were not statistically significant and the source data did not contain assessor-confidence ratings.	**Directional proxy support; not empirically confirmed**

Hypothesis	Proposed relationship	Main finding	Assessment
H7	Institutional AI-guidance clarity is positively associated with disclosure consistency.	Xia et al.'s review of 32 studies supports the need for clearer assessment policies, responsible GenAI use and integrity. However, guidance clarity and disclosure consistency were not jointly measured or statistically tested	**Theoretically and directionally supported**

5.6 Robustness and alternative explanations

The findings should be interpreted as a structured assessment of evidential fit, not as a meta-analysis or causal test. The source studies differ in design, population, measures and outcomes, and no common participant-level dataset includes all constructs. Student–assessor differences may reflect invisible reasoning, differing definitions or classification error; performance associations may reflect prior ability or motivation; and process traces may document activity without establishing understanding. These rival explanations are retained because the framework is intended to guide direct future testing rather than convert adjacent evidence into unwarranted confirmation.

6. Discussion

6.1 Principal findings

This study set out to explore a question that is becoming increasingly difficult for universities to avoid: when students use generative AI, how can educators judge whether the work still represents the student's own understanding and capability? The findings suggest that the answer does not lie in treating all AI use as either acceptable or unacceptable. What matters

more is the way students engage with the technology and whether they remain responsible for the thinking, evaluation and decisions reflected in their work.

The evidence considered across H1–H4 points towards a meaningful distinction between using GenAI as an intellectual aid and using it as a substitute for thought. Students who question AI-generated responses, check evidence, compare alternatives and adapt ideas to the assessment context are more likely to demonstrate cognitive engagement and strategic synthesis. In contrast, simply transferring generated material into an assignment provides little assurance that the student understands or can defend it. Oliveira et al.'s DRIVE framework supports this emphasis on process by showing how student–GenAI interactions can provide evidence of learning that may not be visible in the final written product alone.

H5 brings these relationships together by proposing that engagement mode influences authenticity confidence through a sequence of cognitive engagement, strategic synthesis and stronger authenticity evidence. H6 adds an important qualification: sophisticated reasoning is more convincing when several forms of evidence connect that reasoning to the student. The complete pathway has not yet been tested using one participant-level dataset, so these findings should not be presented as statistical confirmation. Even so, the available evidence offers a credible explanation of how responsible AI engagement may contribute to confidence in the authenticity of student work.

6.2 Interpretation through the conceptual framework

A central contribution of the study is therefore the concept of the **Authenticity Warrant**. In practice, authenticity should not depend on a single prompt record, disclosure statement, detection score or oral response. A more defensible judgement emerges when different sources of evidence tell a broadly consistent story about the student's contribution. Drafts, selected AI interactions, source checks, revision decisions, reflective explanations and oral defence can

collectively show whether the student understood the problem, exercised judgement and remained intellectually responsible for the final submission.

6.3 Theoretical implications

The framework reframes authenticity as an attribution judgement supported by evidence rather than as an intrinsic property of a document. It also separates engagement frequency from cognitive engagement and distinguishes strategic synthesis from the fluency of the final product. The Authenticity Warrant extends this logic by explaining why sophisticated output should contribute more strongly to assessor confidence when the student's process, explanation, disclosure and related performance converge.

6.4 Implications for assessment design

Assessment should make consequential reasoning visible. Suitable approaches include staged submissions, short decision logs, selective prompt-and-response records, source-verification notes, reflective explanations, oral defence and transfer tasks. These elements should be proportionate to the learning outcomes and should be used together where stakes are high; no single artefact should be treated as proof of ownership.

6.5 Implications for assessor literacy

Assessors need the capability to distinguish polished output from demonstrated reasoning, interpret AI disclosures consistently and recognise the limitations of detection tools. They also need shared criteria for evaluating process evidence and for deciding when further inquiry is justified. This is an assessment-literacy requirement, not merely a technology-literacy requirement.

6.6 Institutional and policy implications

Institutional guidance also matters. Students are unlikely to disclose their AI use consistently when expectations differ between programmes, modules or individual assessors. Universities should clearly explain what uses are permitted, what must be declared and what evidence may be requested. This is consistent with Xia et al.'s argument that higher education institutions need to reconsider assessment policies and curricula in response to GenAI. Disclosure should not, however, be treated as proof of learning. It establishes transparency; the quality and authenticity of learning must still be assessed through evidence of understanding, reasoning and judgement.

The findings also reinforce the limitations of automated detection. Turnitin acknowledges that its AI-writing model may not always be accurate and advises that its output should not be used as the sole basis for adverse action against a student. A detection score may justify further inquiry, but it cannot determine authorship or misconduct by itself. Fair academic decisions require wider consideration of the assignment, the student's working process, institutional policy and the student's opportunity to explain the work.

The implications extend beyond universities. In business and human resource management, employees will increasingly be expected to work with AI while remaining accountable for their recommendations. Producing a polished document is not the same as exercising sound professional judgement. A capable practitioner must still identify weak assumptions, verify evidence, consider alternatives and decide whether an AI-generated recommendation is appropriate for the organisation concerned. This is particularly important in SMEs, where recommendations designed for larger organisations may be unrealistic because of financial, technological or managerial constraints.

6.7 Limitations

The paper is limited by its reliance on heterogeneous published studies and aggregate evidence. The selected sources do not measure all constructs using common definitions, and the evidence cannot establish causal direction, serial mediation or moderation. The dashboards operationalise the framework but do not constitute independent empirical tests. Generalisability may also vary by discipline, language background, assessment format and institutional policy, while rapid changes in GenAI capabilities may alter the relevance of particular interaction patterns or detection technologies.

6.8 Future research

Future research should test the complete framework using matched student-level data, validated multi-item measures and assessor judgements collected independently of the student process evidence. Longitudinal and cross-disciplinary studies could examine whether engagement modes change over time and whether authenticity warrants operate similarly across assessment formats and cultural settings. Experimental or quasi-experimental work should also compare assessment designs that vary in the visibility of reasoning, while qualitative studies could investigate how students and assessors interpret fairness, disclosure and intellectual ownership.

7. Conclusion

7.1 Answer to the research problem

Student work authenticity in a GenAI-enabled environment should not be determined by asking whether AI was used. It should be judged by whether the available evidence shows that the student understood the problem, evaluated the material, exercised disciplinary judgement and can account for the decisions represented in the submission.

7.2 Core contribution

The paper's central contribution is the Authenticity Warrant: a convergent evidential basis for attributing demonstrated capability to the student. Process traceability, reasoning evidence, disclosure consistency and corroborating performance are individually incomplete, but together they can support a more defensible judgement than textual inference or an AI-detection score alone.

7.3 Closing implication

Ultimately, authentic work in an AI-enabled environment does not require students or professionals to avoid AI. It requires them to retain ownership of the reasoning and responsibility for the decisions made with its support. The more useful question is therefore not simply *"Was AI used?"* but *"What did the individual understand, evaluate, change and decide?"* By combining AI literacy, clear institutional guidance, transparent disclosure, process-oriented assessment and proportionate authenticity evidence, universities can protect academic standards while preparing graduates for the realities of responsible professional practice.

Acknowledgements

No acknowledgements were provided in the source manuscript.

Disclosure Statement

No potential conflict of interest was reported in the source manuscript.

Funding

No funding information was provided in the source manuscript.

Data Availability Statement

This conceptual study uses evidence reported in the cited publications. No new participant-level dataset was generated or analyzed. The dashboards reproduce or synthesize published aggregate evidence as described in the Method section.

Ethics Statement

The study analyses published aggregate evidence and does not involve the collection or analysis of identifiable human-participant data.

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Appendix A. Evidence Dashboard Links

Sl No	Hypotheses	Link to the Dashboard
1	H1: Engagement mode → cognitive engagement	H1 - Cognitive Engagement Dashboard
2	H2: Cognitive engagement → strategic synthesis	H2 - Strategic Synthesis Dashboard
3	H3: Strategic synthesis → authenticity confidence	H3 Evidence Test Dashboard
4	H4: Engagement mode → authenticity confidence	H4 Authenticity Confidence Dashboard
5	H5: Serial mediation	H5: Serial Mediation Dashboard
6	H6: Authenticity Warrant moderation	H6: Authenticity Warrant Moderation Dashboard
7	H7: Guidance clarity → disclosure consistency	H7 Guidance Clarity disclosure dashboard